The LSST Data Release Production Pipeline, Explained

LSSTCam DRP as defined in `pipelines/LSSTCam/DRP.yaml`

drp_pipe – LSST Science Pipelines documentation

October 18, 2026

Abstract

This document is a guided, end-to-end walkthrough of the Vera C. Rubin Observatory's *Data Release Production* (DRP) pipeline as it is declared for the LSST Camera (LSSTCam) in this repository. Starting from the entry-point `pipelines/LSSTCam/DRP.yaml`, the pipeline chains together more than one hundred *PipelineTasks* drawn from roughly two dozen packages of the LSST Science Pipelines. The pipeline is organised as four *stages* – single-visit processing, recalibration, coaddition, and variability measurement – each subdivided into a sequence of small *steps* that the Butler runs with appropriate sharding (per detector, per visit, per tract, per HEALPix pixel, or globally). For every step we explain what it does, the principal tasks involved, the inputs it consumes, and the data products it produces.

Contents

1 How to read this document 3

2 Architectural prerequisites: Butler, dimensions, quanta 4

3 LSSTCam-specific overrides at a glance 4

4 Stage 1 – Single-visit processing 5
- 4.1 Step 1a – Per-detector single-visit production . . . 5
  - 4.1.1 **isr** – Instrument Signature Removal . . . 5
  - 4.1.2 **calibrateImage** – Initial PSF, WCS and zero point . . . 6
  - 4.1.3 **standardizeSingleVisitStar** . . . 6
  - 4.1.4 Per-detector analysis tasks . . . 7
- 4.2 Step 1b – Per-visit consolidation . . . 7
- 4.3 Step 1c – Tract-level star association . . . 7
  - 4.3.1 **associateIsolatedStar** . . . 7
- 4.4 Step 1d – Survey-wide consolidation of single-visit metadata . . . 8

5 Stage 2 – Recalibration 8
- 5.1 Step 2a – Global photometric calibration with FGCM . . . 8
- 5.2 Step 2b/2c – Astrometric calibration with gbdes . . . 8
- 5.3 Step 2d – Per-visit recalibration . . . 9
- 5.4 Step 2e – Tract-level recalibrated analysis . . . 10

5.5 Step 2f – Survey-wide recalibrated metadata . . . 10

6 Stage 3 – Coaddition and object detection 10
6.1 Step 3a – Per-tract coaddition, detection and measurement . . . 10
6.1.1 Visit selection . . . 10
6.1.2 Warping . . . 11
6.1.3 Coadd assembly . . . 11
6.1.4 Detection and deblending . . . 11
6.1.5 Measurement . . . 12
6.1.6 Galaxy modelling with MultiProFit . . . 12
6.1.7 Object-table assembly . . . 12
6.1.8 Weak-lensing shear with metadetection . . . 13
6.1.9 Per-tract object analysis and image QA . . . 13
6.2 Step 3b – Survey-wide coadd analysis . . . 13
6.3 Steps 3c & 3d – Single-band greyscale and HiPS . . . 14

7 Stage 4 – Variability measurement 14
7.1 Step 4a – Per-visit variability production . . . 14
7.1.1 Visit-image reprocessing . . . 14
7.1.2 Difference image analysis . . . 14
7.1.3 Real-bogus classification . . . 15
7.1.4 Standardisation and consolidation . . . 15
7.1.5 Sky correction (visit mosaic) . . . 15
7.1.6 Visit-level analysis tasks . . . 15
7.2 Step 4b – Solar-System ephemerides . . . 16
7.3 Step 4c – Per-tract variability assembly . . . 16
7.3.1 DIA-source association . . . 16
7.3.2 Forced photometry on objects and DIA objects . . . 16
7.3.3 Source-association analysis . . . 17
7.3.4 Pretty colour-image coadds . . . 17
7.4 Step 4d – Survey-wide variability summaries . . . 17
7.5 Steps 4e & 4f – Cosmetic RGB and EPO HiPS . . . 17

8 Key high-level configurations 18
8.1 ISR: pixel-level corrections . . . 18
8.2 CalibrateImage: pixel calibration deferred to recalibration . . . 18
8.3 Visit summary updates: which provider wins . . . 19
8.4 Coadds: cell coadds are the canonical product . . . 19
8.5 Visit selection: seeing and effective-time cuts . . . 19
8.6 Pretty-coadd cosmetics . . . 19
8.7 Object detection on coadds . . . 20
8.8 Sky correction (focal-plane background) . . . 20
8.9 Solar System: HG12 phase function . . . 20
8.10 DIA schema specialisation . . . 20
8.11 Configuration cheat-sheet . . . 21

9 Measurement plugins applied to sources and objects 21
9.1 Plugins applied by **calibrateImage** . . . 21

9.2 Plugins applied by **measureObjectUnforced** . . . 22
9.3 Plugins applied by **measureObjectForced** . . . 23
9.4 Forced photometry at visit level (**forcedPhotObjectDetector**) . . . 24
9.5 Plugins applied by **detectAndMeasureDiaSource** . . . 24
9.6 MultiProFit galaxy models . . . 24

10 Schema overview: the Object and Source tables **25**
10.1 Object table schema . . . 25
10.1.1 Band-independent identification and geometry . . . 25
10.1.2 Deblending bookkeeping (one set per object) . . . 25
10.1.3 Per-band measurement columns . . . 25
10.1.4 Multi-band MultiProFit Sérsic columns (*not* per-band prefixed) . . . 26
10.1.5 Shape and weak-lensing . . . 26
10.2 Source table schema . . . 26
10.3 Other related tables, at a glance . . . 27

11 Pipeline contracts: how consistency is enforced **28**

12 End-to-end data flow summary **29**

13 Where to learn more **29**

A Task reference **30**

1 How to read this document

The LSSTCam DRP entry point is tiny on purpose:

Listing 1: `pipelines/LSSTCam/DRP.yaml`

```
description: LSSTCam specialization of the DRP pipeline
instrument: lsst.obs.lsst.LsstCam
imports:
  - location: $DRP_PIPE_DIR/pipelines/_ingredients/LSSTCam/DRP.yaml
```

That file in turn imports `pipelines/_ingredients/LSSTCam/DRP.yaml`, which imports the camera-agnostic `base-v2.yaml`, which finally imports the collection of *analysis* ingredient pipelines (visit, coadd, DIA, HiPS, MultiProFit . . .). Together those files declare:

- the `tasks` that comprise the pipeline (Python classes plus overrides for connections and config fields);
- `subsets` – named groups of tasks that can be run together (e.g. `step3a-coadd-tracts`);
- `steps` – the canonical execution order, each with the Butler *dimensions* (`visit`, `detector`, `tract`, . . .) over which it should be sharded;
- `contracts` – declarative consistency checks that the Butler validates before any quanta are generated (for example, "`makeDirectWarp.border` must be $\geq$ the PSF-matching kernel half-size in `makePsfMatchedWarp`").

At a glance

The LSSTCam pipeline is the v2 "base" DRP with a handful of LSSTCam-specific overrides: `IsrTaskLSST` with amp-offset correction enabled, cell-based coadds turned on (`use_cell_coadds: true`), a healpix-based astrometric WCS as the authoritative WCS in **updateVisitSummary**, the LSSTCam DIA schema, and a tightened seeing cut for template-coadd selection. The rest of this document describes the full pipeline these LSSTCam-specific bits plug into.

The remainder of this document is organised *in execution order*. Each major section corresponds to one of the four DRP stages; within each stage we walk through the `step1a, step1b, ...` subsets exactly in the order they appear in the YAML. For every task we list its fully-qualified Python class, a short description of what it does to the data, and – where helpful – a code listing of its YAML stanza from this repository.

2 Architectural prerequisites: Butler, dimensions, quanta

The DRP pipeline is expressed in the *generation-3* ("gen3") Butler framework. Three concepts recur throughout this document and are worth fixing in advance:

Datasets and storage classes
Every input and output of the pipeline is a typed *dataset* (e.g. post_isr_image, preliminary_visit_image, deep_coadd, object, dia_source). The Butler stores them in a *data repository* and retrieves them on demand. Many task configs in the YAML rebind their connection names (e.g. `connections.outputExposure: post_isr_image`) to match this dataset taxonomy.

Dimensions
A dataset is identified by a tuple of dimensions: typical ones are `instrument`, `visit`, `detector`, `tract`, `patch`, `band`, and `healpix3`. Each *step* declares the dimensions over which it should be sharded; for example `step1a-single-visit-detectors` runs `[visit, detector]`, while `step3a-coadd-tracts` runs over `[tract]` and explodes internally into per-patch and per-band quanta.

Quanta
A *quantum* is a single concrete invocation of a *PipelineTask*: e.g. "run `isr` for visit v, detector d". The Butler builds a *QuantumGraph* from the pipeline definition and runs it through the production system (`BPS`). All inter-task dependencies in this document map to edges of that graph.

3 LSSTCam-specific overrides at a glance

`pipelines/_ingredients/LSSTCam/DRP.yaml` customises the base pipeline with the following changes (see the file for the full YAML):

- **isr** is configured to apply LSSTCam's *amp-offset correction* (`ampOffset.doApplyAmpOffset: true`) on top of ordinary instrument signature removal.
- **updateVisitSummary** uses a global FGCM photometric calibration and a healpix-3 astrometric solution as its WCS provider, with the background carried over from the initial summary.
- Coadds use *cell coadds* as the canonical product (`parameters.use_cell_coadds: true`), and **assembleCellCoadd**/**makeDirectWarp** are configured with `numberOfNoiseRealizations: 1`

for metadetection weak lensing.

- Template-coadd visit selection tightens the seeing cut (`maxPsfFwhm: 1.4`) and requires at least 12 visits with $\geq 75\%$ effective-time zeropoint scale; deep-coadd selection allows up to `1.7`.
- DIA tables (**standardizeDiaSource**, **associateDiaSource**, **consolidateDiaSource**, **consolidateDiaObject**) are driven by an LSSTCam-specific schema file (`lsstcam.yaml`).
- **makePrettyDirectWarp** excludes a long list of "problem" detectors so the cosmetic colour image release product is generated only from healthy chips.
- **consolidateSsTables** uses tuned defaults for the HG12 Solar-System phase function fit (`hg12FixedG12: 0.5`, `hg12MagSigmaFloor: 0.05`, `hg12NSigmaClip: 10`).
- Three "recalibrated-star vs. reference catalogue" analysis tasks (`analyzeRecalibratedStar`, `makeAnalysisRecalibratedStarPhotometricRefMatch`, and `analyzeRecalibratedStarPhotometricRefMatch`) and the whole tract-level recalibration subset `step2b-recalibrate-tracts` are excluded for LSSTCam because they are not needed when the camera relies on the healpix-3 global astrometric fit instead.

4 Stage 1 – Single-visit processing

Stage 1 converts raw exposures into calibrated single-detector images and an initial per-visit catalogue of stars. Everything in this stage is causally independent across visits, so it can be massively parallelised.

4.1 Step 1a – Per-detector single-visit production

Sharding: `[visit, detector]`.

The step turns a raw amplifier-level exposure into a calibrated single-detector image together with an initial PSF, WCS, photometric zero point, and a brighter-star catalogue.

4.1.1 isr – Instrument Signature Removal

Class: `lsst.ip.isr.IsrTaskLSST`.

isr reads a raw amplifier image plus the relevant calibration products from the Butler (bias, dark, defects, brighter–fatter kernel, crosstalk coefficients, linearity, CTI, fringes, transmission curves, sky-flat or flat-field) and produces a clean, gain-corrected detector exposure in electrons. The LSSTCam variant of ISR performs, in sequence:

1. digitisation dithering to avoid quantisation bias;
2. serial and parallel overscan subtraction;
3. masking of saturated and suspect pixels;
4. gain scaling (ADU $\rightarrow e^-$);
5. crosstalk corrections (parallel-overscan and between amplifiers);
6. linearity correction;

7. serial charge-transfer-inefficiency (CTI) correction;
8. image assembly and trimming of overscan regions;
9. bias and dark subtraction;
10. brighter–fatter correction;
11. defect masking and interpolation;
12. flat-field (or illumination) correction;
13. fringe removal and transmission-curve attachment;
14. the LSSTCam *amp-offset correction*, which removes the constant additive level differences between amplifier segments that survive flat-fielding.

Output dataset: post_isr_image.

4.1.2 calibrateImage – Initial PSF, WCS and zero point

Class: `lsst.pipe.tasks.calibrateImage.CalibrateImageTask`.

calibrateImage is the single-visit characterisation step: it takes a post_isr_image and produces a science-ready preliminary_visit_image together with an initial PSF model, aperture corrections, an astrometric solution (WCS), a photometric zero point, summary statistics, and a catalogue of the bright stars that were used for calibration. Internally the task:

1. installs a simple Gaussian PSF as a starting point;
2. performs iterative detect–measure–repair on bright stars, improving the PSF model each pass (`install_simple_psf` → `psf_detection` → `psf_subtract_background` → `psf_repair` → `_compute_psf`);
3. measures aperture corrections;
4. matches a curated subset of PSF stars against an external astrometric reference catalogue (the LSSTCam override allows up to 800-pixel offsets in the initial match because butler visit regions have already been pre-refined) and fits a per-detector WCS;
5. matches against a photometric reference catalogue and fits a per-detector photometric zero point;
6. writes out per-detector summary statistics into the exposure metadata.

Output datasets: preliminary_visit_image, preliminary_visit_mask, preliminary_visit_image_background, single_visit_psf_star, single_visit_star_unstandardized, and single_visit_star_schema.

4.1.3 standardizeSingleVisitStar

Class: `lsst.pipe.tasks.postprocess.TransformSourceTableTask`.

Re-shapes the raw afw `Source` catalog from **calibrateImage** into the canonical "pre-source" Parquet schema defined by `schemas/PreSource.yaml`. The output is single_visit_star_detector – a flat tabular catalogue of isolated stars, one row per source.

4.1.4 Per-detector analysis tasks

Three `analysis_tools`-based tasks run alongside production to generate metrics and diagnostic plots:

- **analyzeAmpOffsetMetadata** – summarises the size of the amp-offset corrections that were just applied.
- **analyzeAmpInterfaceOffsetMetadata** – examines the amp–amp interface offsets in post_isr_image metadata.
- **analyzeCalibrateImageMetadata** – harvests metrics from the **calibrateImage** task's metadata stream.
- **analyzeInitialSummaryStats** – runs the `CalexpSummaryAnalysisTask` over each preliminary_visit_image.summaryStats, producing the initial seeing, sky-background and zero-point metrics.

4.2 Step 1b – Per-visit consolidation

Sharding: `[visit]`.

consolidateSingleVisitStar (`ConsolidateSourceTableTask`) gathers all per-detector single_visit_star_detector Parquet tables into a single visit-level table single_visit_star.

consolidateVisitSummary (`ConsolidateVisitSummaryTask`) reads the per-detector preliminary_visit_image headers, PSFs, WCSes, photo-calibs and summary stats and writes them as an `afw.table.ExposureCatalog` called preliminary_visit_summary. This object is the single authoritative "visit metadata" carrier consumed throughout the rest of the pipeline.

The remaining four tasks in this subset are analysis tasks that match the per-visit star table to astrometric and photometric reference catalogues and produce QA plots and metrics: **analyzeSingleVisitStar**, **makeAnalysisSingleVisitStarAstrometricRefMatch** → **analyzeSingleVisitStarAstrometricRefMatch**, **makeAnalysisSingleVisitStarPhotometricRefMatch** → **analyzeSingleVisitStarPhotometricRefMatch**.

4.3 Step 1c – Tract-level star association

Sharding: `[tract]` (skymap constraint mandatory).

4.3.1 associateIsolatedStar

Class: `lsst.pipe.tasks.isolatedStarAssociation.IsolatedStarAssociationTask`.

For every tract in the skymap, this task spatially matches the per-visit single_visit_star observations into multi-band, multi-epoch *isolated stars*. It (i) selects a primary band, (ii) identifies primary-band stars with no nearby neighbours, (iii) cross-matches sources in the other bands to those primaries, and (iv) generates a unique star identifier from a hash of tract and primary star position.

Output datasets: isolated_star_association, isolated_star.

These catalogues are the bedrock of all downstream multi-epoch calibration (FGCM and gbdes both consume them).

analyzeSingleVisitStarAssociation subsequently runs metric plots on the association results.

4.4 Step 1d – Survey-wide consolidation of single-visit metadata

Sharding: global.

makeInitialVisitDetectorTable (`MakeCcdVisitTableTask`) and **makeInitialVisitTable** (`MakeVisitTableTask`) collapse all of the per-visit summary catalogs into two top-level Parquet tables – preliminary_visit_detector_table and preliminary_visit_table – containing one row per detector-visit and per visit, respectively.

The companion analysis tasks **makeAnalysisSingleVisitStarAssociationMetricTable** and **makeAnalysisSingleVisitStarAssociationWholeSkyPlot** consolidate tract-level metrics into a full-survey metric table and a whole-sky plot.

5 Stage 2 – Recalibration

Stage 2 uses the multi-epoch information assembled in step 1c/1d to perform global photometric calibration, a global/tract astrometric solution, and per-visit re-characterisation of the PSF model. The output is a new, more accurate visit_summary and an updated per-visit star catalogue.

5.1 Step 2a – Global photometric calibration with FGCM

Sharding: global.

The Forward Global Calibration Method (FGCM; Burke, Rykoff et al. 2018) builds and minimises a global photometric model that forward-models the atmosphere and instrument – including chromatic effects – to convert instrumental fluxes into top-of-the-atmosphere standard fluxes. The DRP plugs FGCM in via three sequential tasks:

1. **fgcmBuildFromIsolatedStar** reads the isolated_star_association and isolated_star outputs and packages them, together with the preliminary_visit_summary, into the internal data structures the FGCM fitter expects;
2. **fgcmFitCycle** with `doMultipleCycles: true` runs the fitter through the full sequence of fit cycles (the underlying `fgcm` library performs many tens of fit iterations per cycle, with outlier rejection tightened between cycles);
3. **fgcmOutputProducts** writes the resulting per-detector, per-visit photometric calibrations (fgcm PhotoCalibCatalog) and the global zero-point solution that **updateVisitSummary** will consume below.

5.2 Step 2b/2c – Astrometric calibration with gbdes

Sharding: `[tract]` for 2b, `[healpix3]` for 2c.

The gbdes astrometric solver (Bernstein et al. 2017, used here via `lsst.drp.tasks.gbdesAstrometricFit`) fits a hierarchy of polynomial mappings – per-detector and per-visit – that align all observations of every isolated star with each other and with an external reference catalogue. The DRP runs the fit twice with two different geometries:

gbdesAstrometricFit
(step 2b, `tract` dimension): the classic per-tract fit using single_visit_star catalogues. Proper-motion fitting is enabled (`fitProperMotion: true`). *This step is excluded from the LSSTCam*

configuration – LSSTCam relies on the healpix-3 fit instead – but we mention it because it is the canonical interface and is still used in other DRP variants.

gbdesHealpix3AstrometricFit
(step 2c, `healpix3`): fits all visits intersecting each level-3 HEALPix pixel using isolated stars (`useIsolatedStars: true`), again with proper motions enabled. This is the WCS provider chosen for LSSTCam in **updateVisitSummary**.

Output dataset: gbdesAstrometricFitSkyWcsCatalog (or its healpix-3 sibling) – one fitted WCS per detector-visit, stored in an `ExposureCatalog`.

5.3 Step 2d – Per-visit recalibration

Sharding: `[visit]`.

This is where the per-detector single-visit data are stitched together into the final, calibrated per-visit products. The tasks run in this order:

1. **refitPsfModelDetector** (`FinalizeCharacterizationDetectorTask`) re-fits the PSF model and aperture corrections on each detector using the full, multi-epoch isolated star sample (isolated_star), instead of the per-visit sources used in step 1a. It writes refit_psf_star_detector (the calibration stars used) and refit_psf_models_detector (the new PSF/ap-corr model).

2. **consolidateRefitPsfModelDetector** merges the per-detector tables into per-visit refit_psf_star and refit_psf_models.

3. **gaussianProcessesTurbulenceFit** (`lsst.drp.tasks.fit_turbulence.GaussianProcessesTurbulenceFitTask`) fits a Gaussian-process model of atmospheric turbulence residuals in the WCS using the isolated stars, providing a spatially-correlated correction on top of the gbdes fit.

4. **updateVisitSummary** (`lsst.drp.tasks.update_visit_summary.UpdateVisitSummaryTask`) composes a new visit_summary `ExposureCatalog` by overriding fields of preliminary_visit_summary with the new PSF/ap-corr/WCS/photo-calib information:

   - PSF and aperture-correction overrides come from refit_psf_models;
   - the WCS comes from the healpix-3 gbdes fit (`wcs_provider: healpix3`);
   - the photo-calib comes from FGCM (`photo_calib_provider: global`);
   - the background is reused from preliminary_visit_image_background (`background_provider: input_summary`).

   A configurable `wcs_consistency_threshold` ensures that any one detector's WCS may not differ from its neighbours by more than 0.5 pixels.

5. **recalibrateSingleVisitStar** (`WriteRecalibratedSourceTableTask`) re-photometers the per-visit star catalogue using the new visit_summary and writes recalibrated_star_unstandardized.

6. **standardizeRecalibratedStar** and **consolidateRecalibratedStar** project that into a flat, canonical Parquet table recalibrated_star.

7. Analysis tasks **analyzeRecalibratedStar**, **makeAnalysisRecalibratedStarAstrometricRefMatch** and **analyzeRecalibratedStarAstrometricRefMatch** regenerate the diagnostic plots from step 1b

using the now-recalibrated catalogues. (For LSSTCam the photometric-reference variants and **analyzeRecalibratedStar** are excluded.)

5.4 Step 2e – Tract-level recalibrated analysis

Sharding: `[tract]`.

analyzeRecalibratedStarAssociation repeats the star-association plots of step 1c but using recalibrated_star, providing the final QA of the calibration solution.

fitStellarMotion (`lsst.drp.tasks.fit_stellar_motion.FitStellarMotionTask`) re-fits proper motions for each isolated star using the recalibrated positions, producing the multi-epoch astrometric solution used by downstream forced-photometry tasks.

5.5 Step 2f – Survey-wide recalibrated metadata

Sharding: global.

makeVisitDetectorTable and **makeVisitTable** are the recalibrated analogs of step 1d: they collapse the new visit_summary `ExposureCatalog`s into the survey-level Parquet tables visit_detector_table and visit_table. Two final analysis tasks (**makeAnalysisRecalibratedStarAssociationMetricTable**, **makeAnalysisRecalibratedStarAssociationWholeSkyPlot**) assemble whole-sky QA plots.

Note

In LSSTCam, the `step2b-recalibrate-tracts` subset is excluded entirely. The pipeline relies on the healpix-3 fit (`2c`) plus the turbulence Gaussian process (`2d`) for its astrometric solution.

6 Stage 3 – Coaddition and object detection

Stage 3 combines visits into deep image stacks (*coadds*), detects astronomical objects on them, deblends overlapping sources, measures them in all bands, and produces the object catalogue.

6.1 Step 3a – Per-tract coaddition, detection and measurement

Sharding: `[tract]` (skymap constraint mandatory). This step contains the bulk of the production tasks; we walk through it in groups.

6.1.1 Visit selection

selectDeepCoaddVisits (`BestSeeingSelectVisitsTask`) and **selectTemplateCoaddVisits** (`BestSeeingQuantileSelectVisitsTask`) read all visit_summary catalogues intersecting the tract and choose two visit subsets:

- the *deep* subset for science coadds – LSSTCam allows seeing up to 1.7″ and requires the effective time zero-point scale to be ≥ 0.75;
- the *template* subset for difference imaging – a tighter 1.4″ seeing cut, at least 12 visits, again with the zero-point quality cut.

6.1.2 Warping

makeDirectWarp (`MakeDirectWarpTask`) re-projects each selected preliminary_visit_image onto the tract's standard pixel grid without altering the per-visit PSF. It also generates direct_warp_masked_fraction (the fraction of each pixel masked) and – because LSSTCam runs metadetection downstream – $N = 1$ noise realisations per warp (`numberOfNoiseRealizations: 1`).

makePsfMatchedWarp (`MakePsfMatchedWarpTask`) additionally convolves each warp with a per-visit kernel that matches the visit's PSF to a uniform target PSF, producing the psf_matched_warp that the `CompareWarp` algorithm uses for artefact detection.

6.1.3 Coadd assembly

Three coadd flavours are produced in parallel from the same warps:

assembleDeepCoadd (`CompareWarpAssembleCoaddTask`)
Builds the deep science coadd by online-mean stacking, with artefact rejection by comparing each direct warp against its PSF-matched twin. Outputs deep_coadd_predetection (the coadd image before detection), an artefact-mask catalogue, deep_coadd_n_image (per-pixel input visit count), and an inputMap if cell coadds are disabled.

assembleTemplateCoadd (`CompareWarpAssembleCoaddTask`)
Same algorithm with a tighter visit selection produces the template_coadd used downstream for difference imaging.

assembleCellCoadd (`AssembleCellCoaddTask`)
The *cell coadd* alternative: each tract is divided into small cells with overlapping inner/outer bounding boxes; within each cell, only visits whose PSFs are fully sampled across the cell are stacked, guaranteeing a continuous, well-defined coadd PSF. For LSSTCam this is the canonical coadd (`use_cell_coadds: true`). Output: deep_coadd_cell_predetection, plus an inputMap.

makeDeepCoaddInputSummaryTract and **makeTemplateCoaddInputSummaryTract** summarise, for each tract, which visits contributed to each coadd.

makeHealSparsePropertyMaps (`HealSparsePropertyMapTask`) consumes the deep-coadd input map plus the visit_summary catalogs and produces *HealSparse* survey property maps – one HEALPix sparse map per property per band. The standard set includes total exposure time, observation epochs (min/max/mean), PSF size and ellipticity (mean, weighted mean), magnitude limits, sky background level and noise, and differential chromatic refraction shifts.

6.1.4 Detection and deblending

detectCoaddPeaks (`DetectCoaddSourcesTask`) runs an SDSS-like detection over each band's deep coadd (or cell coadd), producing per-band footprints and peaks. For LSSTCam the configuration enables "temp wide" background-aware detection (`doTempWideBackground: true`, `tempWideBackground.binSize: 128`) and re-estimates the background after detection at the same binning, with sky-object placement targeted at 400 sources per patch. *Output:* object_detection (per-band).

mergeObjectDetection (`MergeDetectionsTask`) combines the per-band detections into a single multi-band footprint catalogue (object_detection_merged), eliminating duplicates and preserving every band's peak list.

deconvolve (`lsst.meas.extensions.scarlet.deconvolveExposureTask.DeconvolveExposureTask`) performs an approximate deconvolution of the deep coadd against its own PSF, producing deconvolved_deep_coadd which improves the quality of subsequent deblending.

deblendCoaddFootprints (`DeblendCoaddSourcesMultiTask`) runs the *Scarlet* multi-band deblender on the merged detection footprints. Scarlet models each blend as a sum of source-shaped, spectrally-coherent components, fitted simultaneously across all bands. *Outputs:* object_deblend (a per-band catalogue with deblended children), object_scarlet_models (the source models themselves), and object_deblend_schema.

6.1.5 Measurement

measureObjectUnforced (`MeasureMergedCoaddSourcesTask`) measures every deblended source in each band's coadd *independently*, using the deblender output to redistribute flux among blend children. The output object_unforced_measurement contains adaptive moments, aperture fluxes, PSF fluxes, Gaussian fluxes, and a host of `base_*` flags.

mergeObjectMeasurement (`MergeMeasurementsTask`) picks a reference band for each object based on its multi-band measurements (a *which band has the best S/N* decision) and writes object_ref_measurement.

measureObjectForced (`ForcedPhotCoaddTask`) re-measures every object in every band using the reference-band centroid and shape from object_ref_measurement, producing *forced* fluxes (object_forced_measurement). Forced photometry across bands is what makes the LSST object catalogue colour-consistent.

6.1.6 Galaxy modelling with MultiProFit

The MultiProFit code (DMTN-312) fits parametric Gaussian-mixture galaxy models simultaneously across all bands. Three tasks run in sequence:

1. **fitDeepCoaddPsfGaussians** (`MultiProFitCoaddPsfFitTask`) fits a Gaussian-mixture representation of the deep-coadd PSF at each object position.
2. **fitDeblendedObjectsExp** (`MultiProFitCoaddExpFitTask`) fits an exponential-disk model to each deblended object in every band.
3. **fitDeblendedObjectsSersic** (`MultiProFitCoaddSersicFitTask`) fits a free-index Sérsic model.

The outputs (object_psf_multiprofit, object_exp_multiprofit, object_sersic_multiprofit) are folded into the standardised object table below.

6.1.7 Object-table assembly

rewriteObject (`WriteObjectTableTask`) collates the unforced, forced and MultiProFit-PSF measurements into a single multi-band object_unstandardized Parquet file per patch.

standardizeObject (`TransformObjectCatalogTask`) projects that into the canonical object_patch schema, merging in the MultiProFit Exp/Sersic fits and applying the official functor definitions.

consolidateObject (`ConsolidateObjectTableTask`) gathers all patches in a tract into the tract-level object_all Parquet table.

computeObjectEpochs (`ComputeObjectEpochsTask`) attaches, per object, the mean epoch of the input visits that contributed to its deep coadd pixels.

consolidateParentTable (`ConsolidateParentTractTask`) builds a parent-only sibling table object_parent.

splitPrimaryObject (`SplitPrimaryTask`) splits object_all into two non-overlapping tables: object (the primary detection at each sky location) and object_nonprimary (the duplicates that arose from tract overlap). Only object is intended for science.

6.1.8 Weak-lensing shear with metadetection

metadetectionShear (`MetadetectionShearTask`) runs the *metadetection* weak-lensing measurement on the cell coadds: it artificially shears the coadd by small amounts, re-detects and re-measures sources at each shear, and from those finite differences derives a calibrated shear response per object. The metadetection algorithm achieves $< 0.1\%$ shear bias even in the presence of realistic blending, which is the LSST Dark Energy requirement. *Output:* object_shear_patch.

consolidateShearObject gathers per-patch shear tables into the tract-level object_shear_all.

6.1.9 Per-tract object analysis and image QA

A long list of analysis tasks run after the object table is built:

- **analyzeObjectTableCore**, **analyzeObjectParentTableCore** – canonical plot/metric bundles on object_all and object_parent.
- **catalogMatchTract** → **refCatObjectTract** – match object_all to an astrometric reference and produce astrometric residual diagnostics.
- **photometricCatalogMatch** → **photometricRefCatObjectTract** – the photometric equivalent.
- **plotPropertyMapTract** – per-tract visualisation of the HealSparse property maps.
- Binning and "whole tract image" tasks: **makeBinnedCoaddImage** and **makeWholeTractImage** for the deep coadd; **makeBinnedTemplateNImage** and **makeWholeTractTemplateNImage** for the template N-image; **makeBinnedDeepNImage** and **makeWholeTractDeepNImage** for the deep N-image.
- **validateObjectTableCore** – column-validation checks on the assembled object table.

6.2 Step 3b – Survey-wide coadd analysis

Sharding: global.

consolidateHealSparsePropertyMaps (`ConsolidateHealSparsePropertyMapTask`) combines all per-tract HealSparse maps into a single survey-wide map per property.

makeDeepCoaddInputSummary and **makeTemplateCoaddInputSummary** aggregate the tract-level "which-visits-went-where" summaries into survey-wide tables.

The remaining tasks roll the tract-level analysis-tool metrics up into whole-sky plots and the survey-wide object table summary: **analyzeObjectTableSurveyCore**, **makeMetricTableObjectTableCore**, **objectTableCoreWholeSkyPlot**, **makeMetricTableObjectParentTableCore**, **objectParent**

TableCoreWholeSkyPlot, **makeMetricTableObjectTableCoreRefCatMatch**, **objectTableCoreRefCatMatchWholeSkyPlot**, and **plotPropertyMapSurvey**.

6.3 Steps 3c & 3d – Single-band greyscale and HiPS

Sharding: global.

Step 3c (**makeSingleBandRGBCoadd***) generates a stretched greyscale visualisation of each band's deep coadd (u, g, r, i, z, y).

Step 3d builds Hierarchical Progressive Surveys (HiPS) maps per band. For each band the pipeline runs

- **makeHighOrderHips*** – generate the finest-resolution HiPS tiles using HEALPix-based tessellation;
- **makeLowOrderHips*** – coarsen those into all the intermediate levels;
- **writeAllSkyHipsInfo*** – write the HiPS metadata that web clients (Aladin Lite, the Rubin Science Platform) need to consume the tiles.

HiPS is the IVOA-standard format for serving multiresolution all-sky imagery; combined with the HealSparse property maps it forms the visual "slippy map" surface of the data release.

7 Stage 4 – Variability measurement

Stage 4 returns to the visit level and uses the deep object catalogue and template coadd as priors. It produces difference images, transient detections (dia_source), associated dia_object catalogues, forced photometry on every object at every visit, ephemerides for known Solar System objects, and a cosmetic colour image release product.

7.1 Step 4a – Per-visit variability production

Sharding: `[visit]`.

7.1.1 Visit-image reprocessing

reprocessVisitImage (`lsst.drp.tasks.reprocess_visit_image.ReprocessVisitImageTask`) rebuilds a final, science-ready single-visit image (visit_image) by applying the refit PSF, the new WCS, FGCM photo-calib, and updated background to post_isr_image. The preliminary mask is consumed and an updated mask is written. The task also produces a new per-detector source_unstandardized catalogue.

standardizeSource and **consolidateSource** project that into the canonical source_detector and source_all schemas. **splitPrimarySource** produces the primary source table (one row per primary detection per visit).

7.1.2 Difference image analysis

The classical Alard & Lupton (1998) image-subtraction pipeline is implemented as a chain of four PipelineTasks:

1. **buildTemplate** (`GetTemplateTask`) carves the relevant patch of the global template_coadd into the single-visit detector footprint and re-projects it onto the visit's pixel grid, producing templat e_detector.

2. **subtractImages** (`AlardLuptonSubtractTask`) computes a matching kernel that PSF-matches the template to the science image, convolves the template, and subtracts. The output difference_ image_predetection is the decorrelated difference image (the `DecorrelateALKernelTask` subtask undoes the noise correlations introduced by the convolution). A template_matched image and the kernel itself are also saved. Crucially, the task can fall back to its own source detection if the input source list is insufficient (`allowKernelSourceDetection: true`).

3. **detectAndMeasureDiaSource** (`DetectAndMeasureTask`) detects positive and negative excursions in the difference image at S/N ≥ 5 and measures them. "Sky sources" – random positions in clean sky – are also placed for photometric noise characterisation. Outputs: dia_source_unfiltere d (the catalogue), difference_image (post-measurement difference), difference_streaks (a streak mask), and the schema.

4. **filterDiaSource** (`FilterDiaSourceCatalogTask`) applies preliminary flag-based cuts (long-trailed source detection, sky-source removal) producing dia_source_unstandardized, long_trailed_ dia_source, and rejected_dia_source.

7.1.3 Real-bogus classification

computeReliability (`RBTransiNetTask`) runs the TransiNet PyTorch CNN ("*RBTransiNet*") on the science/template/difference image triplet at each DIA source position to assign a *reliability* score in $[0, 1]$. Published TransiNet benchmarks reach $\sim 96\%$ classification accuracy on the image triplet. *Output:* dia_source_reliability.

filterDiaSourcePostReliability (`FilterDiaSourceReliabilityTask`) splits sources by the reliability score into dia_source_high_reliability (kept) and dia_source_low_reliability (rejected).

7.1.4 Standardisation and consolidation

standardizeDiaSource (`TransformDiaSourceCatalogTask`) reads the LSSTCam DIA schema (`lsstcam.yaml`) and projects dia_source_high_reliability into the canonical dia_source_detector flat schema.

consolidateVisitDiaSource gathers per-detector DIA sources into the per-visit dia_source_visit.

7.1.5 Sky correction (visit mosaic)

skyCorr (`SkyCorrectionTask`) performs a focal-plane-wide background fit using all detectors in a visit, producing the preliminary_visit_image_skyCorr_visit_mosaic and a sky mosaic background. The tuning uses two background scales (an 8192-px coarse model and a 4096-px smooth model on top), masking objects before fitting. *Note:* a TODO in the YAML (`DM-53870`) flags that this should eventually move back to stage 2.

7.1.6 Visit-level analysis tasks

analyzeRecalibratedStarObjectMatch matches the per-visit recalibrated stars against the object table. **analyzeSource** produces metric plots on the source catalogue. **makeAnalysisSourceAstromet**

ricRefMatch → **analyzeSourceAstrometricRefMatch** and **makeAnalysisSourcePhotometricRefMatch** → **analyzeSourcePhotometricRefMatch** reproduce the "against-the-refcat" metric/plot pattern used in stages 1 and 2.

7.2 Step 4b – Solar-System ephemerides

Sharding: global.

generateEphemerides (`GenerateEphemeridesTask`) reads preliminary_visit_table and, for every visit, queries the Minor Planet Center (or local MPC mirror) to obtain the predicted positions of every known Solar System object that should be in the field of view. These ephemerides are the inputs that **associateDiaSource** downstream uses to attribute matched DIA detections to known SSObjects.

Note

The YAML notes that this step's inputs are ready after step 1c; it is positioned here only to run as late as possible for DP2. A future refactor (DM-53596) will move it into stage 2.

7.3 Step 4c – Per-tract variability assembly

Sharding: `[tract]`.

7.3.1 DIA-source association

associateDiaSource (`DrpAssociationPipeTask`) takes all per-detector dia_source_detector catalogues in the tract, plus the SSO ephemerides, and:

1. links repeated DIA detections at the same sky position into dia_object_pre_calc;
2. attaches each detection to its associated DIA object, producing dia_source_patch;
3. associates DIA detections that match the predicted ephemeris of a known Solar System object into ss_source_associated;
4. flags ephemerides with no match (ss_object_unassociated).

calculateDiaObject (`DrpDiaCalculationPipeTask`) computes the per-object summary statistics that go into the final dia_object_patch (lightcurve moments, periodicity, colour information, . . .).

consolidateDiaSource and **consolidateDiaObject** (`ConsolidateTractTask`) merge patches into the tract-level dia_source and dia_object Parquet tables under the LSSTCam DIA schema.

7.3.2 Forced photometry on objects and DIA objects

forcedPhotObjectDetector (`ForcedPhotDetectorTask`) measures forced fluxes on every visit_image at every object_patch position ("forced light curves of every object"). Its outputs are standardised by **standardizeObjectForcedSource** into object_forced_source_all, then split by **splitPrimaryObjectForcedSource** into the primary object_forced_source and object_forced_source_nonprimary.

forcedPhotDiaObjectDetector repeats the procedure but uses the DIA-object positions (`refCat: dia_object_patch`) as the reference catalogue. **standardizeDiaObjectForcedSource** projects the result into dia_object_forced_source.

7.3.3 Source-association analysis

associateAnalysisSource, **analyzeSourceAssociation**, and **analyzeDiaSourceTableTract** produce per-tract diagnostic plots and metrics on the assembled source and dia_source catalogues.

7.3.4 Pretty colour-image coadds

A second pass of coaddition runs here, intended only for the public cosmetic colour-image release product (*not* for science). For LSSTCam, **makePrettyDirectWarp** excludes a long list of known-bad detectors so the colour image is only built from healthy chips. The rest of the chain is:

1. **makePrettyDirectWarp** – warps each visit_image with sky correction applied (`doApplySkyCorr: true`) onto the tract grid.
2. **makePrettyPsfMatchedWarp** – the PSF-matched companion.
3. **assemblePrettyCoadd** (`CompareWarpAssembleCoaddTask`) – coadds the warps with the SPIKE mask plane suppressed and without writing artefact masks.
4. **makeBinnedPrettyCoaddImage** → **makeWholeTractPrettyCoaddImage** – binned visualisation.
5. **makeBinnedPrettyNImage** → **makeWholeTractPrettyNImage** – the matching N-image.

7.4 Step 4d – Survey-wide variability summaries

Sharding: global.

makeAnalysisSourceAssociationMetricTable and **makeAnalysisSourceAssociationWholeSkyPlot** consolidate the tract-level source-association analyses into survey-wide tables and plots.

consolidateSsTables (`ConsolidateSsTablesTask`) is the headline Solar-System deliverable: it folds the per-visit ss_source_associated catalogues into the official ss_source (one row per detection) and ss_object (one row per Solar System body) tables. The LSSTCam configuration uses a fixed-G_{12} HG12 phase function for the brightness model (`hg12FixedG12: 0.5`), with 0.05-mag noise floor and 10-σ outlier rejection.

7.5 Steps 4e & 4f – Cosmetic RGB and EPO HiPS

Sharding: global.

Step 4e applies last-mile cosmetic fixes to the pretty coadds before making the public colour image:

- **fixBackgroundPrettyCoadd** – removes residual backgrounds;
- **fixStarsPrettyCoadd** – repairs star cores that have been saturated;
- **makeRGBPicturePrettyCoaddGRI** – combines the g, r, i bands into a colour image.

Step 4f turns those colour coadds into HiPS for outreach (**makeHighOrderHipsEpoGRI**, **makeLowOrderHipsEpoGRI**, **writeAllSkyHipsInfoEpoGRI**) and the canonical multi-band HiPS for the public data release (**makeHighOrderHipsGRI**, **makeLowOrderHipsGRI**, **writeAllSkyHipsInfoGRI**).

8 Key high-level configurations

Beyond which tasks are present, the YAML and the `config/` Python files set a number of *high-level* configuration knobs that materially change the science output. The single most important "you need to know this" fact for each major task is summarised below. The full table at the end of this section gives a quick-reference index.

8.1 ISR: pixel-level corrections

isr is configured with the LSSTCam-specific `IsrTaskLSST` class. The main correction toggles on the LSSTCam configuration are:

- **Amp-offset corrections are turned ON** – `ampOffset.doApplyAmpOffset: true`. This is the LSSTCam-specific override; for ImSim (`LSSTCam-imSim/DRP.yaml`) the offsets are *computed* (`doAmpOffset: true`) but *not applied* (`ampOffset.doApplyAmpOffset: false`), because that camera's simulated flats already match across amplifier boundaries.
- Brighter–fatter correction, defect masking, crosstalk, linearity, CTI, fringe and transmission-curve attachment are all on by default in `IsrTaskLSST` and remain so for LSSTCam.
- The output is rebound to post_isr_image via `connections.outputExposure`, so the canonical Butler dataset name is dataset-stable across the pipeline.

8.2 CalibrateImage: pixel calibration deferred to recalibration

calibrateImage is configured via `config/calibrateImage.py`. The single most consequential choice is:

Listing 2: `config/calibrateImage.py` (excerpt)

```
config.do_calibrate_pixels = False
```

This says "do not bake the photometric zero point or backgrounds into the pixels yet" – the pixels stay in instrumental electron units, and only the preliminary_visit_image's `photoCalib` object carries the calibration. The reason given in the file comment is that "in DRP not calibrating the pixels (or the backgrounds) until we have our final photometric calibration makes everything downstream much simpler." Stage 2 calibration products are baked in only by **updateVisitSummary** and **reprocessVisitImage**.

The same config file also tunes:

- **Aperture radii.** The `base_CircularApertureFlux` plugin uses the canonical DRP radii $[3, 6, 9, 12, 17, 25, 35, 50, 70]$ pixels, with sinc-weighting only out to $R \leq 12$ pixels (`maxSincRadius = 12.0`).
- **FGCM-friendly compensated tophat fluxes.** `base_CompensatedTophatFlux` is computed with `apertures: [12, 17]` pixels – one inner and one outer aperture – to give FGCM the inner/outer aperture pair plus a local-background estimate it expects.
- **Star selection is fully data-driven.** `star_selector["science"].doUnresolved = False` and `doSignalToNoise = False` – this defers the "is this a star" decision to the downstream isolated-star association and PSF-refit machinery, instead of cutting on unresolved/SNR thresholds at the visit level.

- **LSSTCam astrometric matcher tolerance.** The LSSTCam YAML loosens `astrometry.matcher.maxOffsetPix=800` and `astrometry_ref_loader.pixelMargin=800`. These assume that Butler visit regions have been pre-refined; they are explicitly *not* appropriate for the nightly-validation or quickLook variants.

8.3 Visit summary updates: which provider wins

updateVisitSummary is the meeting-point of FGCM, gbdes, and the per-visit PSF refit. The provider matrix for LSSTCam is:

Field	Provider	Configured by
PSF model	refit per-detector + visit consolidation	`refit_psf_models`
Aperture corrections	same source as PSF	`refit_psf_models`
WCS	healpix-3 gbdes fit	`wcs_provider:healpix3`
Photo-calib	global FGCM	`photo_calib_provider:global`
Background	carried over from preliminary	`background_provider:input_summary`

A consistency check (`wcs_consistency_threshold: 0.5`) caps the difference between any one detector's healpix-3 WCS and its neighbours at half a pixel.

8.4 Coadds: cell coadds are the canonical product

The single highest-level coadd choice is `parameters.use_cell_coadds: true`. That one Boolean propagates everywhere via the pipeline's `contracts` (see Section 11), so:

- **detectCoaddPeaks**, **deblendCoaddFootprints**, **measureObjectUnforced**, **measureObjectForced**, **fitDeepCoaddPsfGaussians**, **fitDeblendedObjectsSersic** and **deconvolve** all switch to their cell-coadd code paths;
- **makeDirectWarp** writes one noise realisation per warp (`numberOfNoiseRealizations: 1`) so that **metadetectionShear** has the correlated noise field it needs;
- **assembleCellCoadd** also writes a single noise realisation (`num_noise_realizations: 1`).

8.5 Visit selection: seeing and effective-time cuts

The LSSTCam YAML overrides the default visit selectors as follows:

Selector	Cuts (LSSTCam)
selectDeepCoaddVisits	PSFFWHM $\leq 1.7''$; `effTimeZeroPointScale` ≥ 0.75.
selectTemplateCoaddVisits	PSFFWHM $\leq 1.4''$; `effTimeZeroPointScale` ≥ 0.75; at least 12 visits.

8.6 Pretty-coadd cosmetics

The pretty-coadd colour-image path is configured via `config/prettyPicture.py`. Highlights:

- `config.luminanceConfig.stretch=400` and `midtone=0.2` set the luminance stretch.
- Three exposure brackets (`exposureBrackets=[1.1,1,0.8]`) are HDR-combined.
- `doPsfDeconvolve=False` – no PSF deconvolution for the colour image; the deconvolution is reserved for the science path via **deconvolve**.

- Local-contrast enhancement is on with `clarity=1.3` and a 2-level wavelet pyramid.
- `gamutMapperConfig.gamutMethod="inpaint"` reduces clipping in saturated star cores; `colorConfig.saturation=0.65` keeps colours physically plausible.
- **makePrettyDirectWarp** additionally *excludes 53 specific LSSTCam detectors* from the cosmetic warp (the long `select.excludeDetectors` list in the YAML).

8.7 Object detection on coadds

detectCoaddPeaks (LSSTCam) uses these key overrides:

Config	Value / meaning
`detection.skyObjects.nSources`	400 (sky-source target per patch)
`detection.doTempWideBackground`	`true` (model wide-scale background before detection)
`detection.tempWideBackground.binSize`	128 px
`detection.tempWideBackground.useApprox`	`false`
`detection.reEstimateBackground`	`true` (refit local background after first detection)
`detection.background.binSize`	128 px
`detection.background.useApprox`	`false`

8.8 Sky correction (focal-plane background)

skyCorr fits two cascaded focal-plane background models:

- `bgModel1`: an 81.92×81.92 mm tile size at 0.01 mm/pixel, i.e. an 8192 × 8192-pixel coarse scale, with `doSmooth:false` – this captures the long-wavelength scattered-light pattern.
- `bgModel2`: 40.96×40.96 mm (4096 × 4096-pixel) with `doSmooth:true` and `smoothScale:1.0` – a smaller-scale residual model.
- `doSky:true` – subtract the modelled sky after masking detected sources via the `maskObjects.detection` configuration (`thresholdType:stdev`, `isotropicGrow:false`).

8.9 Solar System: HG12 phase function

consolidateSsTables fits an HG12 magnitude–phase function to each Solar System object. LSSTCam pins:

- `hg12FixedG12:0.5` – the G_{12} slope parameter is held fixed rather than fit per object;
- `hg12MagSigmaFloor:0.05` – a 0.05 mag noise floor is added in quadrature to every photometric uncertainty;
- `hg12NSigmaClip:10` – 10 σ outliers are clipped from the fit.

8.10 DIA schema specialisation

Four LSSTCam DIA tasks (**standardizeDiaSource**, **associateDiaSource**, **consolidateDiaSource**, **consolidateDiaObject**) all set `doUseSchema:True` with `schemaFile:lsstcam.yaml`. That schema file in the `sdm_schemas` package is the authoritative column definition for the LSSTCam-flavoured DIA tables; among other things it sets per-column units, types and SDM (Science Data Model)

documentation entries that downstream consumers (the Rubin Science Platform's TAP service) use to render the catalogue. The `parameters.schemaFile` pipeline parameter in the LSSTCam `DRP.yaml` ensures all four tasks see the same schema.

8.11 Configuration cheat-sheet

Knob	LSSTCam DRP setting
`ampOffset.doApplyAmpOffset` (**isr**)	`true` – *apply* amp-offset
`do_calibrate_pixels` (**calibrateImage**)	`false` – defer to stage 2
`wcs_provider` (**updateVisitSummary**)	`healpix3` (gbdes healpix-3 fit)
`photo_calib_provider`	`global` (FGCM)
`background_provider`	`input_summary` (preliminary)
`parameters.use_cell_coadds`	`true`
`makeDirectWarp.numberOfNoiseRealizations`	1
`assembleCellCoadd.num_noise_realizations`	1
`selectDeepCoaddVisits.maxPsfFwhm`	$1.7''$
`selectTemplateCoaddVisits.maxPsfFwhm`	$1.4''$ ($+ \geq 12$ visits)
`detectCoaddPeaks.skyObjects.nSources`	400 per patch
`detection.background.binSize`	128 px
`skyCorrbgModel1/bgModel2sizes`	8192 / 4096 px
`hg12FixedG12` (**consolidateSsTables**)	0.5
`schemaFile` (DIA)	`lsstcam.yaml`

9 Measurement plugins applied to sources and objects

LSST's measurement framework is built around *plugins* (small algorithm modules from `lsst.meas.base` and its extensions) that the per-source measurement tasks run in a fixed dependency order: centroids first (execution order 0.0), then shapes (1.0), then fluxes (2.0). The exact plugin set differs at the visit (Source) and coadd (Object) measurement stages.

9.1 Plugins applied by calibrateImage

calibrateImage runs three measurement sub-tasks on a single visit image:

`psf_source_measurement`
: Measures sources during the iterative PSF-determination loop. The default plugin set is augmented by `ext_shapeHSM_HigherOrderMomentsSource`, which provides higher-order HSM moments used by FGCM and downstream PSF QA.

`star_detection`
: Detects bright stars with a threshold multiplier of 1.0 (`includeThresholdMultiplier:1.0`) – equivalent to the canonical DRP detection threshold.

`star_measurement`
: The main bright-star measurement, which writes the single_visit_star catalogue.

The `star_measurement` plugin set is, in execution order:

Plugin	What it measures
`base_PixelFlags`	Standard pixel-level flag set (bad, saturated, cosmic ray, edge, interpolated, . . .).
`base_SdssCentroid`	Adaptive centroid from the SDSS algorithm.
`base_SdssShape`	Adaptive Gaussian-weighted moments (I_{xx}, I_{yy}, I_{xy}).
`base_PsfFlux`	PSF-weighted flux fit, the headline point-source flux.
`base_GaussianFlux`	Flux integrated over the best-fit Gaussian footprint.
`base_CircularApertureFlux`	Aperture fluxes at radii $[3, 6, 9, 12, 17, 25, 35, 50, 70]$ px, sinc-weighted up to 12 px.
`base_CompensatedTophatFlux`	Compensated tophat at 12 & 17 px for FGCM.
`base_FPPosition`	Focal-plane position of the centroid.
`base_LocalBackground`	Background level in an annulus around the source.
`base_Variance`	Variance plane value at the centroid.
`base_Blendedness`	Fractional flux contamination from neighbours (per Bosch et al. 2018, §4.9.11).
`base_Jacobian`	Astrometric Jacobian determinant at the source position (for sky-area-weighted fluxes).
`ext_shapeHSM_HsmPsfMomentsDebiased`	PSF moments with synthetic noise added to match the source S/N.
`ext_shapeHSM_HsmShapeRegauss`	Hirata & Seljak (2003) "Regaussianization" PSF-corrected shear estimator.
`ext_shapeHSM_HigherOrderMomentsSource`	Third- and fourth-order moments of source and PSF, used in shape-systematics QA.

The HSM (Hirata–Seljak–Mandelbaum) family in `lsst.meas.extensions.shapeHSM` bridges to the GalSim library and provides the precision shear estimators that the metadetection weak-lensing path consumes.

9.2 Plugins applied by measureObjectUnforced

The unforced coadd measurement pass runs roughly the same plugin family, but on the deep coadd or cell coadd and on *every deblended child* from Scarlet (not just bright stars). The default `MeasureMergedCoaddSourcesTask` plugin set comprises:

Plugin	What it measures
base_PixelFlags	Pixel-level flags (same as above).
base_SdssCentroid	Adaptive centroid for measurement.
base_SdssShape	Adaptive moments; produces shape_xx, shape_yy, shape_xy in the object table.
base_PsfFlux	Forced and free PSF flux.
base_GaussianFlux	Best-fit Gaussian flux.
base_CircularApertureFlux	Aperture fluxes at the canonical radii.
base_LocalBackground	Local background annulus.
base_Variance	Variance at centroid.
base_Blendedness	Blendedness from neighbour contamination.
base_FootprintArea	Number of pixels in the detection footprint.
base_ClassificationExtendedness	PSF-vs-model flux ratio; the extendedness column.
base_ClassificationSizeExtendedness	Moments-based extendedness (the sizeExtendedness column).
base_NormalizedCompensatedTophatFlux	Normalised compensated tophat for FGCM.
modelfit_CModel	CModel: a joint exponential + de Vaucouleurs galaxy fit, yielding the cModelFlux, bdE1/E2, bdReB/ReD, bdFluxB/D columns.
ext_gaap_GaapFlux	Gaussian-aperture-and-PSF fluxes (gaapPsfFlux, gaap0p7Flux, gaap1p0Flux, gaap1p5Flux, gaap2p5Flux, gaap3p0Flux, gaapOptimalFlux).
ext_photometryKron_KronFlux	Kron radius and elliptical Kron flux.
ext_shapeHSM_HsmPsfMoments	PSF moments at the object position.
ext_shapeHSM_HsmPsfMomentsDebiased	Debiased PSF moments.
ext_shapeHSM_HsmShapeRegauss	PSF-corrected Regaussianization shear – the source of the hsmShapeRegauss_e1/e2/sigma columns.
ext_shapeHSM_HigherOrderMoments*	3rd and 4th higher-order moments on both source and PSF (the hsm_moments_30/21/12/03/40/31/22/13/04 columns).
ext_shapeHSM_HsmSourceMomentsRound	Round-Gaussian source moments (ixxRound/iyyRound/ixyRound).

The Scarlet deblender output is consumed by these plugins in one of two modes (controlled by measureObjectUnforced.doConserveFlux): either using the deblender's flux-conservation reapportionment (the default), or using the Scarlet models directly.

9.3 Plugins applied by measureObjectForced

The forced coadd pass re-runs measurement using the reference-band centroid and shape as fixed priors. Centroid/shape plugins are therefore replaced by their Transformed* variants:

Plugin	What it measures
base_TransformedCentroid	Carry the reference-band centroid into this band.
base_TransformedShape	Carry the reference-band shape into this band.
base_PsfFlux	PSF flux at the fixed centroid – the headline forced psfFlux.
base_CircularApertureFlux	Forced aperture fluxes at the canonical radii.
base_LocalBackground	Local background.
base_PixelFlags, base_Variance	Flags and variance at the fixed position.
modelfit_CModel	Forced CModel; produces cModelFlux per band.
ext_gaap_GaapFlux	Forced GAaP fluxes.
ext_photometryKron_KronFlux	Forced Kron flux.

9.4 Forced photometry at visit level (forcedPhotObjectDetector)

For per-visit forced light-curves of every object, the plugin set is deliberately minimal – the work to measure a single fixed centroid in a single image is small:

- `base_TransformedCentroid`, `base_TransformedShape` from the reference catalogue;
- `base_PsfFlux` and `base_PsfFlux_apCorr`;
- `base_PixelFlags` for pixel-level flags;
- `base_LocalBackground`.

The same plugin set is used by **forcedPhotDiaObjectDetector**, swapped only to use dia_object_patch as the reference catalogue.

9.5 Plugins applied by detectAndMeasureDiaSource

DIA-source measurement on the difference image uses a tailored plugin set designed for transients (compact, possibly negative flux). Notable plugins are:

- `base_NaiveCentroid` and `base_SdssCentroid` – centroid the detection.
- `base_SdssShape` – adaptive moments.
- `base_PsfFlux` – the headline DIA flux, with explicit support for negative-flux sources.
- `ext_trailedSources_NaiveTrailedSource` – identify and fit linearly-trailed sources (asteroid streaks) so they can be filtered out downstream by **filterDiaSource**.
- `base_SkySource` – placement and measurement of "sky" control sources in clean background, used by **filterDiaSource** and downstream noise QA.
- `base_PixelFlags` – pixel-level flags, including difference-image-specific ones such as `streak` and `injected`.

9.6 MultiProFit galaxy models

Three MultiProFit tasks add explicit parametric galaxy models on top of the measurement plugin pipeline:

- **fitDeepCoaddPsfGaussians** fits a Gaussian-mixture approximation to each band's coadd PSF at the object position. This becomes the PSF input for the galaxy fits.
- **fitDeblendedObjectsExp** fits a single-band-by-band exponential-disk profile model. The model is a sum of Gaussian mixtures whose total profile is the Gaussian-mixture approximation of an exponential (Sérsic $n = 1$).
- **fitDeblendedObjectsSersic** fits a free-index Sérsic profile. The model is again a Gaussian-mixture approximation (DMTN-312 documents the exact mixture); the free index n becomes the `sersic_index` column of the Object table.

All three operate row-by-row on single objects and consolidate their per-patch outputs into tract-level Parquet catalogues that **standardizeObject** merges into the canonical Object table.

10 Schema overview: the Object and Source tables

The two principal tabular outputs of the DRP – the object table (per-tract, per-patch deblended sources measured on coadds) and the source table (per-visit detections measured on single visit images) – are written by **standardizeObject** and **standardizeSource** respectively, using *functor* schema files from `pipe_tasks/schemas`. Their canonical user-facing column lists are defined in the Rubin Science Data Model (`sdm_schemas`). This section gives a high-level overview of both; for the full per-column definitions see the SDM schema publication at https://sdm-schemas.lsst.io/dp1.html.

10.1 Object table schema

The Object table contains *static measurements of astronomical objects* detected and measured on the deep coadds. It is one row per deblended object per tract, with no parent objects (only deblended children), and duplicates across tract boundaries are removed by **splitPrimaryObject**. As released in DP1 the table has 1,296 columns; the LSSTCam DRP produces an equivalent schema. Columns fall into the following groups.

10.1.1 Band-independent identification and geometry

Column	Type	Description
`objectId`	long	Unique object ID (primary key).
`parentObjectId`	long	Parent ID from the reference-band deblend.
`tract`, `patch`	long	Skymap location.
`coord_ra`, `coord_dec`	double	Fiducial ICRS centroid (degrees) used for indexing.
`coord_raErr`, `coord_decErr`, `coord_ra_dec_Cov`	float	1-σ centroid errors and covariance.
`refBand`	char	Reference band picked by **mergeObjectMeasurement**.
`x`, `y`	double	SDSS centroid in reference-band pixel coordinates.
`shape_xx/yy/xy`	float	Reference-band adaptive moments.
`refExtendedness`	float	PSF-vs-model flux extendedness (0=star, 1=galaxy).
`refSizeExtendedness`	float	Moments-based extendedness.
`footprintArea`	int	Pixels in the reference-band footprint.
`ebv`	float	Galactic E(B–V) at the centroid (Schlegel et al. 1998).

10.1.2 Deblending bookkeeping (one set per object)

`detect_isIsolated`, `detect_fromBlend`, `detect_isDeblendedModelSource`, `deblend_nChild`, `deblend_nPeaks`, `deblend_failed`, `deblend_skipped`, `deblend_isolatedParent`, `deblend_parentTooBig`, `deblend_tooManyPeaks`, `deblend_masked`, `deblend_incompleteData`, `deblend_iterations`, `deblend_peak_center_x`, `deblend_peak_center_y`, `deblend_logL`.

10.1.3 Per-band measurement columns

For each of the six LSSTCam bands $b \in \{u, g, r, i, z, y\}$, the table carries a parallel set of columns prefixed with `b_`. The most important groups are:

Pattern	Source
b_ra, b_dec, b_raErr, b_decErr, b_ra_dec_Cov	Per-band astrometric centroid (forced).
b_psfFlux, b_psfFluxErr, b_psfMag, b_psfMagErr	base_PsfFlux, forced on coadd in band b.
b_free_psfFlux, b_free_psfFluxErr, b_free_psfFlux_flag	Same, but unforced (band-only).
b_gaap{Psf,0p7,1p0,1p5,2p5,3p0,Optimal}Flux	GAaP fluxes at PSF and 0.7/1.0/1.5/2.5/3.0 arcsec apertures, all PSF-renormalised to a 1.15×-larger target Gaussian PSF.
b_kronRad, b_kronFlux	Kron radius and flux from ext_photometryKron.
b_apNNFlux ($N \in \{03, 06, 09, 12, 17, 25, 35, 50, 70\}$)	Forced circular aperture fluxes at the canonical pixel radii.
b_calibFlux	Duplicate of b_ap12Flux for legacy compatibility.
b_cModelFlux, b_cModelFluxErr, b_cModelMag, b_cModelFlux_inner	Forced CModel galaxy flux (exp + deV decomposition).
b_free_cModelFlux*	Unforced CModel.
b_bdE1, b_bdE2, b_bdReB, b_bdReD, b_bdChi2, b_bdFluxB, b_bdFluxD	CModel bulge/disk decomposition outputs.
b_ixx, b_iyy, b_ixy	HSM source moments.
b_ixxPSF, b_iyyPSF, b_ixyPSF, b_ixxDebiasedPSF, ...	HSM PSF moments, including the noise-debiased variant.
b_ixxRound, b_iyyRound, b_ixyRound	Isotropic-Gaussian-weighted moments.
b_hsmShapeRegauss_e1, b_hsmShapeRegauss_e2, b_hsmShapeRegauss_sigma	Hirata–Seljak Regauss shear and per-component noise.
b_hsm_moments_NN, b_hsm_momentsPsf_NN for $NN \in \{30, 21, 12, 03, 40, 31, 22, 13, 04\}$	Higher-order HSM moments of source and PSF.
b_extendedness, b_sizeExtendedness, b_blendedness	Per-band extendedness and blendedness.
b_pixelFlags_*	Pixel flags: bad, cr, edge, interpolated, nodata, saturated, suspect, plus centre variants.
b_epoch	Mean epoch of the input visits (MJD TAI) computed by **computeObjectEpochs**.
b_inputCount	Number of input visits at this position.

10.1.4 Multi-band MultiProFit Sérsic columns (*not* per-band prefixed)

sersic_x, sersic_y, sersic_ra, sersic_dec, sersic_reff_x, sersic_reff_y, sersic_rho, sersic_index, plus the *Err variants and quality flags (sersic_chisq_reduced, sersic_n_iter, sersic_n_eval_jac, sersic_no_data_flag, sersic_unknown_flag). The per-band MultiProFit fluxes are b_sersicFlux, b_sersicFluxErr for each b.

10.1.5 Shape and weak-lensing

The Object table does not directly carry metadetection shear; that catalogue (object_shear_all) lives alongside the Object table and is keyed by objectId for joining.

10.2 Source table schema

The Source table contains *per-visit detections* measured directly on visit_image, independently of the coadd-based Object catalogue. It is dimensioned by (band, instrument, day_obs, physical_filter,

visit), with one Parquet file per visit. As released in DP1 the table has 156 columns; the LSSTCam DRP produces an equivalent schema. The major column groups are:

Group	Columns
Identification	sourceId (primary key), parentSourceId, visit, detector, band, physical_filter.
Astrometry	coord_ra, coord_dec, ra, dec, raErr, decErr, ra_dec_Cov, x, y, xErr, yErr.
PSF photometry	psfFlux, psfFluxErr, psfFlux_apCorr, psfFlux_apCorrErr, psfFlux_area, psfFlux_flag*.
Gaussian flux	gaussianFlux, gaussianFluxErr, gaussianFlux_flag.
Aperture fluxes	apNNFlux, apNNFluxErr, apNNFlux_flag for $NN \in \{03, 06, 09, 12, 17, 25, 35, 50, 70\}$; also the raw apFlux_12_0_instFlux etc. used by FGCM.
Calibration aperture	calibFlux, calibFluxErr (duplicates of ap12Flux), normCompTophatFlux_instFlux (for FGCM).
Shape	ixx, iyy, ixy, ixxPSF, iyyPSF, ixyPSF, ixxDebiasedPSF, iyyDebiasedPSF, ixyDebiasedPSF (HSM moments of source and PSF).
HSM shear / flags	hsmShapeRegauss_flag*, iDebiasedPSF_flag*, hsmPsfMoments_flag*.
Background	sky, skyErr, localBackground_instFlux, localBackground_instFluxErr, localBackground_flag*.
Variance / Jacobian	variance_value, variance_flag*, jacobian_value, jacobian_flag.
Star / galaxy	extendedness, sizeExtendedness, blendedness_abs, blendedness_flag*.
Footprint	footprintArea_value, invalidPsfFlag.
Pixel flags	pixelFlags_bad, pixelFlags_cr, pixelFlags_crCenter, pixelFlags_edge, pixelFlags_interpolated, pixelFlags_interpolatedCenter, pixelFlags_nodata, pixelFlags_offimage, pixelFlags_saturated, pixelFlags_saturatedCenter, pixelFlags_suspect, pixelFlags_suspectCenter.
Centroid flags	centroid_flag* (including badError, edge, notAtMaximum, resetToPeak, ...).
Calibration provenance	calib_astrometry_used, calib_photometry_used, calib_photometry_reserved, calib_psf_candidate, calib_psf_used, calib_psf_reserved.
Deblending	deblend_nChild, deblend_deblendedAsPsf, deblend_hasStrayFlux, deblend_masked, deblend_parentTooBig, deblend_patchedTemplate, deblend_rampedTemplate, deblend_skipped, deblend_tooManyPeaks.
Sky-source marker	sky_source (true for control sky sources placed by **detectAndMeasureDiaSource**/base_SkySource).

The source table is keyed by visit and is split by **splitPrimarySource** into source (one row per primary detection, used for science) and source_nonprimary (rows on detector overlap, kept only for completeness).

10.3 Other related tables, at a glance

For completeness, the DRP also produces several closely related tables that share the column conventions above:

Table	What it contains
ForcedSource	Forced photometry of every Object at every visit. Keyed by (`objectId`, `visit`, `detector`); columns include forced `psfFlux`, `psfDiffFlux`, `psfFlux_apCorr`, aperture fluxes, local background, and the pixel-flag and `calib_*` provenance flags.
DiaSource	Per-difference-image transient detections. Keyed by `diaSourceId`; carries the science image, template image, difference image, PSF flux, shape, `reliability` (from **computeReliability**), and a `ssObjectId` for matches to Solar System bodies.
DiaObject	Time-aggregated DIA sources: one row per persistent transient, with light-curve summary statistics, periodicity statistics, and colour information.
ForcedSourceOnDiaObject	Forced photometry of every `diaObjectId` at every visit – the canonical lightcurves for variables and transients.
SSObject	One row per known Solar System body, with the HG12 phase-function fit (`H`, `G12`, `Herr`, `nDiaSources`, plus orbital metadata).
SSSource	One row per DIA detection that was associated with a Solar System body; carries the predicted ephemeris and the residual between predicted and detected position.
Visit / CcdVisit	The visit_table and visit_detector_table produced in step 2f – one row per visit and per detector-visit, with PSF, WCS, photo-calib, observing conditions, and `effTimeZeroPointScale`.
CoaddPatches	Per-patch metadata for the deep coadds.

11 Pipeline contracts: how consistency is enforced

Beyond the dataset graph, the pipeline declares a list of `contracts` – assertions that the Butler validates before any quantum graph is built. Notable examples include:

- Connection alignment: e.g. the template coadd produced by **assembleTemplateCoadd** must be the one consumed by **buildTemplate**; the matched template and difference image from **subtractImages** must match the ones consumed by **detectAndMeasureDiaSource**.
- PSF flag propagation: if `makeDirectWarp.useVisitSummaryPsf` is true, then no `calib_psf_*` flags may be propagated into the object table (because those flags would already be baked into the visit-summary PSF).
- Geometric consistency: `makeDirectWarp.border` must be $\geq$ the PSF-matching kernel's half-cell-size; the warp cache size must match the coadd's PsfCoadd cache size; and `makeDirectWarp.numberOfNoiseRealizations` must be at least `assembleCellCoadd.num_noise_realizations`.
- Calibration consistency: when illumination correction is enabled in **calibrateImage**, **isr** must also apply the flat (`doFlat: true`); the warp must use the flat-background ratio; reprocessing must do the same; and **fgcmBuildFromIsolatedStar** must *not* apply the WCS Jacobian (which is already accounted for).
- Cell-coadd consistency: **detectCoaddPeaks**, **deblendCoaddFootprints**, **measureObjectUnforced**, **measureObjectForced**, **fitDeepCoaddPsfGaussians**, **fitDeblendedObjectsSersic**, and **deconvolve** must all agree on whether cell coadds are in use.

These contracts are why the cell-coadd parameter (`use_cell_coadds`) is propagated everywhere from a single top-level value, and why turning it on for LSSTCam is a single-line override.

12 End-to-end data flow summary

The DRP can be summarised by the chain of canonical datasets that flow through it, with the producing task in brackets:

1. Raw exposure $\xrightarrow{\text{isr}}$ post_isr_image $\xrightarrow{\text{calibrateImage}}$ preliminary_visit_image + single_visit_star_unstandardized.
2. Per-detector star catalogs are standardised and consolidated into single_visit_star, and per-detector summaries into preliminary_visit_summary.
3. Tract-level **associateIsolatedStar** builds isolated_star_association and isolated_star.
4. FGCM + gbdes produce fgcmPhotoCalibCatalog and gbdesHealpix3AstrometricFitSkyWcsCatalog; the per-visit PSF is refit on the multi-epoch sample (refit_psf_models); **updateVisitSummary** bakes all of them into the final visit_summary.
5. Per-visit stars are re-photometered against visit_summary into recalibrated_star; stellar motions are fit into the multi-epoch astrometric solution.
6. visit_summary drives warping; warps drive coadd assembly into deep_coadd, deep_coadd_cell_predetection, template_coadd, and the input maps that feed deep_coadd_input_map and the HealSparse property maps.
7. Detection → merge → deconvolve → Scarlet deblend → unforced measurement → ref-band merge → forced measurement → MultiProFit fits → object_all → object + object_nonprimary.
8. Metadetection on cell coadds produces object_shear_all.
9. template_coadd + visit_image drive image differencing → dia_source → dia_object.
10. Forced photometry on visit_image at every object and dia_object position yields object_forced_source and dia_object_forced_source.
11. Solar System ephemerides × DIA-source positions → ss_source and ss_object.
12. The pretty-coadd chain produces the public colour image, which together with the property maps becomes the HiPS layers served by the Rubin Science Platform.

13 Where to learn more

- LSST Science Pipelines API reference: https://pipelines.lsst.io/.
- DP1 processing overview, including pre-rendered pipeline graphs of the same DRP: https://dp1.lsst.io/processing/index.html and https://dp1.lsst.io/processing/summary/index.html.
- `lsst.ip.isr` module documentation: https://pipelines.lsst.io/modules/lsst.ip.isr/index.html.
- `IsrTaskLSST` class reference: https://pipelines.lsst.io/py-api/lsst.ip.isr.IsrTaskLSST.html.

- `CalibrateImageTask`: https://pipelines.lsst.io/py-api/lsst.pipe.tasks.calibrateImage.CalibrateImageTask.html.
- FGCM and `lsst.fgcmcal`: https://pipelines.lsst.io/modules/lsst.fgcmcal/index.html; Burke, Rykoff et al. 2018 for the algorithm.
- `GbdesAstrometricFitTask`: https://pipelines.lsst.io/modules/lsst.drp.tasks/tasks/lsst.drp.tasks.gbdesAstrometricFit.GbdesAstrometricFitTask.html; Bernstein et al. 2017 for the underlying WCS solver.
- Scarlet deblender: https://pipelines.lsst.io/modules/lsst.meas.extensions.scarlet/catalog.html.
- MultiProFit (DMTN-312): https://dmtn-312.lsst.io/.
- Metadetection for Rubin: https://astro.theoj.org/article/75280.
- `AlardLuptonSubtractTask`: https://pipelines.lsst.io/modules/lsst.ip.diffim/tasks/lsst.ip.diffim.AlardLuptonSubtractTask.html.
- RBTransiNet real-bogus classifier: https://pipelines.lsst.io/py-api/lsst.meas.transiNet.RBTransiNetTask.html; DMTN-216 for the algorithm.
- DIA pipeline overview: https://dp1.lsst.io/processing/dia/index.html.
- HealSparse property maps: https://dp1.lsst.io/products/spmaps/index.html.
- HiPS: https://dp1.lsst.io/products/hips/index.html and the IVOA HiPS recommendation https://www.ivoa.net/documents/HiPS/.

A Task reference

The following tables list every PipelineTask declared by the LSSTCam DRP, organised by stage and step. Each row gives the task's label, its fully-qualified Python class, and a one-sentence summary of what it does. The detailed narrative for each task is in the body of this document.

Stage 1 – Single-visit processing

Step 1a – per-detector single-visit production (`[visit, detector]`).

Task	Brief description
isr lsst.ip.isr.IsrTaskLSST	Remove instrument signatures from a raw amplifier image (overscan, bias, dark, gain, crosstalk, linearity, CTI, brighter–fatter, defects, flat, fringe, amp-offset) and write a clean detector image in electrons.
calibrateImage lsst.pipe.tasks.calibrateImage.CalibrateImageTask	Iteratively fit the PSF, aperture corrections, a per-detector WCS, and a photometric zero point on a post-ISR image, and write the preliminary visit image plus a catalogue of bright calibration stars.

Task	Brief description
standardizeSingleVisitStar lsst.pipe.tasks.postprocess.TransformSourceTableTask	Project the raw `afw Source` catalogue from **calibrateImage** into the canonical flat Parquet "PreSource" schema.
analyzeAmpOffsetMetadata lsst.analysis.tools.tasks.DatasetMetadataAnalysisTask	Summarise the size and distribution of the amp-offset corrections applied by **isr**.
analyzeAmpInterfaceOffsetMetadata lsst.analysis.tools.tasks.DatasetMetadataAnalysisTask	Examine the per-amplifier interface offsets recorded in the post_isr_image metadata.
analyzeCalibrateImageMetadata lsst.analysis.tools.tasks.MetadataAnalysisTask	Harvest QA metrics from the metadata stream emitted by **calibrateImage**.
analyzeInitialSummaryStats lsst.analysis.tools.tasks.CalexpSummaryAnalysisTask	Produce diagnostic plots and metrics from each preliminary visit image's summary statistics (seeing, sky, zero point).

Step 1b – per-visit consolidation (`[visit]`).

Task	Brief description
consolidateSingleVisitStar lsst.pipe.tasks.postprocess.ConsolidateSourceTableTask	Gather all per-detector star tables for a visit into one visit-level single_visit_star Parquet table.

Task	Brief description
consolidateVisitSummary lsst.pipe.tasks.postprocess.ConsolidateVisitSummaryTask	Bundle per-detector PSFs, WCSes, photo-calibs and summary stats into the visit-level preliminary_visit_summary ExposureCatalog.
analyzeSingleVisitStar lsst.analysis.tools.tasks.SourceTableVisitAnalysisTask	Run the standard source-table analysis tools on the per-visit star catalogue.
makeAnalysisSingleVisitStarAstrometricRefMatch lsst.analysis.tools.tasks.astrometricCatalogMatch.AstrometricCatalogMatchVisitTask	Match per-visit stars against the astrometric reference catalogue to produce a matched catalogue for QA.
analyzeSingleVisitStarAstrometricRefMatch lsst.analysis.tools.tasks.refCatSourceAnalysis.RefCatSourceAnalysisTask	Plot astrometric residuals (ΔRA, ΔDec) and compute metrics from the matched catalogue above.
makeAnalysisSingleVisitStarPhotometricRefMatch lsst.analysis.tools.tasks.photometricCatalogMatch.PhotometricCatalogMatchVisitTask	Match per-visit stars against the photometric reference catalogue.

Task	Brief description
analyzeSingleVisitStarPhotometricRefMatch lsst.analysis.tools.tasks.refCatSourcePhotometricAnalysis.RefCatSourcePhotometricAnalysisTask	Plot photometric residuals and compute zero-point QA metrics from the photometric match.

Step 1c – tract-level star association ([tract]).

Task	Brief description
associateIsolatedStar lsst.pipe.tasks.isolatedStarAssociation.IsolatedStarAssociationTask	Spatially match the per-visit stars in a tract into multi-band, multi-epoch *isolated stars* – the bedrock catalogue for FGCM and gbdes.
analyzeSingleVisitStarAssociation lsst.analysis.tools.tasks.associatedSourcesTractAnalysisTask	Produce QA plots and metrics from the tract-level isolated-star association.

Step 1d – survey-wide consolidation (global).

Task	Brief description
makeInitialVisitDetectorTable lsst.pipe.tasks.postprocess.MakeCcdVisitTableTask	Collapse all detector-level visit summaries into the survey-wide preliminary_visit_detector_table.

Task	Brief description
makeInitialVisitTable lsst.pipe.tasks.postprocess.MakeVisitTableTask	Collapse the visit-level summaries into the survey-wide preliminary_visit_table.
makeAnalysisSingleVisitStarAssociationMetricTable lsst.analysis.tools.tasks.MakeMetricTableTask	Aggregate the tract-level association metrics into a single survey-wide metric table.
makeAnalysisSingleVisitStarAssociationWholeSkyPlot lsst.analysis.tools.tasks.WholeSkyAnalysisTask	Plot the association metrics across the whole sky on a HEALPix mollweide.

Stage 2 – Recalibration

Step 2a – global photometric calibration with FGCM (global).

Task	Brief description
fgcmBuildFromIsolatedStar lsst.fgcmcal.fgcmBuildFromIsolatedStars.FgcmBuildFromIsolatedStarsTask	Convert the isolated-star catalogues plus the preliminary visit summaries into the input data structures that FGCM expects.
fgcmFitCycle lsst.fgcmcal.fgcmFitCycle.FgcmFitCycleTask	Run the FGCM forward-modelling fitter through multiple fit cycles (atmosphere × instrument → standard fluxes).

Task	Brief description
fgcmOutputProducts lsst.fgcmcal.fgcmOutputProducts.FgcmOutputProductsTask	Write the per-visit, per-detector photo-calibrations and global zero-point solution to the Butler.

Steps 2b & 2c – astrometric calibration with gbdes ([tract] / [healpix3]).

Task	Brief description
gbdesAstrometricFit lsst.drp.tasks.gbdesAstrometricFit.GbdesAstrometricFitTask	Fit a hierarchical (per-detector × per-visit) polynomial astrometric solution across all visits of a tract, with proper motions. *Excluded for LSSTCam.*
gbdesHealpix3AstrometricFit lsst.drp.tasks.gbdesAstrometricFit.GbdesAstrometricFitTask	Same fit but anchored on level-3 HEALPix pixels using the isolated-star catalogue; this is the WCS provider for LSSTCam.

Step 2d – per-visit recalibration ([visit]).

Task	Brief description
refitPsfModelDetector lsst.pipe.tasks.finalizeCharacterization.FinalizeCharacterizationDetectorTask	Re-fit the per-detector PSF and aperture-correction maps using the multi-epoch isolated-star sample.
consolidateRefitPsfModelDetector lsst.pipe.tasks.finalizeCharacterization.ConsolidateFinalizeCharacterizationDetectorTask	Merge the per-detector PSF refits into the visit-level refit_psf_models and refit_psf_star tables.

Task	Brief description
gaussianProcessesTurbulenceFit lsst.drp.tasks.fit_turbulence.GaussianProcessesTurbulenceFitTask	Fit a Gaussian-process model of small-scale atmospheric turbulence residuals on top of the gbdes WCS.
updateVisitSummary lsst.drp.tasks.update_visit_summary.UpdateVisitSummaryTask	Compose the final visit_summary by overriding PSF, ap-corr, WCS, and photo-calib in the preliminary summary with the refit/gbdes/FGCM products.
recalibrateSingleVisitStar lsst.pipe.tasks.postprocess.WriteRecalibratedSourceTableTask	Re-photometer the per-visit star catalogue using the new visit_summary.
standardizeRecalibratedStar lsst.pipe.tasks.postprocess.TransformSourceTableTask	Project recalibrated stars into the canonical per-detector Parquet schema.
consolidateRecalibratedStar lsst.pipe.tasks.postprocess.ConsolidateSourceTableTask	Combine per-detector recalibrated catalogues into the visit-level recalibrated_star.
analyzeRecalibratedStar lsst.analysis.tools.tasks.SourceTableVisitAnalysisTask	Run the standard source-table analysis tools on the recalibrated star catalogue. *Excluded for LSSTCam.*

Task	Brief description
makeAnalysisReca librated StarAstr ometricR efMatch lsst.analysis.tools.tasks.astrometricCatalogMatch.AstrometricCatalogMatchVisitTask	Match recalibrated stars against the astrometric reference catalogue.
analyzeR ecalibra tedStarA strometr icRefMat ch lsst.analysis.tools.tasks.refCatSourceAnalysis.RefCatSourceAnalysisTask	Plot astrometric residuals using the recalibrated star matches.
makeAnalysisReca librated StarPhot ometricR efMatch lsst.analysis.tools.tasks.photometricCatalogMatch.PhotometricCatalogMatchVisitTask	Match recalibrated stars against the photometric reference catalogue. *Excluded for LSSTCam.*
analyzeR ecalibra tedStarP hotometr icRefMat ch lsst.analysis.tools.tasks.refCatSourcePhotometricAnalysis.RefCatSourcePhotometricAnalysisTask	Plot photometric residuals from the recalibrated reference match. *Excluded for LSSTCam.*

Steps 2e & 2f – tract analysis + survey-wide metadata (`[tract]` / global).

Task	Brief description
analyzeRecalibratedStarAssociation lsst.analysis.tools.tasks.AssociatedSourcesTractAnalysisTask	Repeat the star-association QA plots using the recalibrated catalogue, for tract-level diagnostics.
fitStellarMotion lsst.drp.tasks.fit_stellar_motion.FitStellarMotionTask	Re-fit per-star proper motions and parallaxes from the recalibrated multi-epoch positions.
makeVisitDetectorTable lsst.pipe.tasks.postprocess.MakeCcdVisitTableTask	Build the final survey-wide visit_detector_table from the recalibrated visit summaries.
makeVisitTable lsst.pipe.tasks.postprocess.MakeVisitTableTask	Build the final survey-wide visit_table.
makeAnalysisRecalibratedStarAssociationMetricTable lsst.analysis.tools.tasks.MakeMetricTableTask	Aggregate the recalibrated-star association metrics into a survey-wide metric table.
makeAnalysisRecalibratedStarAssociationWholeSkyPlot lsst.analysis.tools.tasks.WholeSkyAnalysisTask	Plot the recalibrated-star association metrics on a full-sky map.

Stage 3 – Coaddition and object detection

Step 3a – per-tract production: warping, coadd assembly, property maps (`[tract]`).

Task	Brief description
selectDeepCoaddVisits lsst.pipe.tasks.selectImages.BestSeeingSelectVisitsTask	Choose visits to include in the deep science coadd (good seeing, decent zero-point scale).
selectTemplateCoaddVisits lsst.pipe.tasks.selectImages.BestSeeingQuantileSelectVisitsTask	Choose the best-seeing visits for the difference-imaging template coadd.
makeDirectWarp lsst.drp.tasks.make_direct_warp.MakeDirectWarpTask	Re-project each selected visit onto the tract's pixel grid without altering its native PSF.
makePsfMatchedWarp lsst.drp.tasks.make_psf_matched_warp.MakePsfMatchedWarpTask	Convolve each direct warp with a per-visit kernel that matches its PSF to a uniform target PSF.
assembleDeepCoadd lsst.drp.tasks.assemble_coadd.CompareWarpAssembleCoaddTask	Online-mean stack the direct warps, comparing to the PSF-matched warps to reject transient artefacts; outputs the deep science coadd.
assembleTemplateCoadd lsst.drp.tasks.assemble_coadd.CompareWarpAssembleCoaddTask	Same algorithm, run with the tighter template visit selection, producing the difference-imaging template_coadd.
assembleCellCoadd lsst.drp.tasks.assemble_cell_coadd.AssembleCellCoaddTask	Build the cell-coadd: stack each spatial cell only from visits whose PSF is fully sampled, guaranteeing a continuous PSF (the canonical LSSTCam coadd).

Task	Brief description
makeDeepCoaddInputSummaryTract lsst.drp.tasks.make_coadd_input_summary.MakeCoaddInputSummaryTract	Summarise which visits contributed which pixels to each tract of the deep coadd.
makeTemplateCoaddInputSummaryTract lsst.drp.tasks.make_coadd_input_summary.MakeCoaddInputSummaryTract	Same summary for the template coadd.
makeHealSparsePropertyMaps lsst.pipe.tasks.healSparseMapping.HealSparsePropertyMapTask	Compute per-tract HealSparse maps of observing properties (exposure time, seeing, sky, DCR, mag-limit, . . .).

Step 3a (continued) – detection, deblending, measurement, object table.

Task	Brief description
detectCoaddPeaks lsst.pipe.tasks.multiBand.DetectCoaddSourcesTask	Run SDSS-like detection on each band's coadd, with temp-wide background handling, to produce per-band peak lists and footprints.
mergeObjectDetection lsst.pipe.tasks.mergeDetections.MergeDetectionsTask	Combine the per-band detections into a single, de-duplicated multi-band footprint catalogue.
deconvolve lsst.meas.extensions.scarlet.deconvolveExposureTask.DeconvolveExposureTask	Produce a Scarlet-quality approximate deconvolution of the deep coadd to improve deblending.

Task	Brief description
deblendCoaddFootprints lsst.pipe.tasks.deblendCoaddSourcesPipeline.DeblendCoaddSourcesMultiTask	Run the multi-band Scarlet deblender on the merged footprints, producing per-source models and deblended children.
measureObjectUnforced lsst.pipe.tasks.multiBand.MeasureMergedCoaddSourcesTask	Measure each deblended source in each band independently (moments, aperture, PSF, Gaussian, Kron fluxes).
mergeObjectMeasurement lsst.pipe.tasks.mergeMeasurements.MergeMeasurementsTask	Pick the best reference band per object from its multi-band measurements.
measureObjectForced lsst.drp.tasks.forcedPhotCoadd.ForcedPhotCoaddTask	Re-measure every object in every band using the reference-band centroid & shape, producing colour-consistent forced fluxes.
fitDeepCoaddPsfGaussians lsst.meas.extensions.multiprofit.pipetasks_fit.MultiProFitCoaddPsfFitTask	Fit a Gaussian-mixture representation of the coadd PSF at each object position (input for galaxy fits).
fitDeblendedObjectsExp lsst.meas.extensions.multiprofit.pipetasks_fit.MultiProFitCoaddExpFitTask	MultiProFit fit of an exponential-disk galaxy model simultaneously across all bands.

Task	Brief description
fitDebledObjectsSersic lsst.meas.extensions.multiprofit.pipetasks_fit.MultiProFitCoaddSersicFitTask	MultiProFit fit of a free-index Sérsic galaxy model across all bands.
rewriteObject lsst.pipe.tasks.postprocess.WriteObjectTableTask	Collate unforced + forced + MultiProFit-PSF measurements into per-patch object_unstandardized Parquet files.
standardizeObject lsst.pipe.tasks.postprocess.TransformObjectCatalogTask	Apply the canonical object functor schema and merge in the MultiProFit Exp & Sérsic columns.
consolidateObject lsst.pipe.tasks.postprocess.ConsolidateObjectTableTask	Combine all patches in a tract into the tract-level object_all.
computeObjectEpochs lsst.drp.tasks.compute_object_epochs.ComputeObjectEpochsTask	Attach the mean epoch of the contributing input visits to each object.
consolidateParentTable lsst.pipe.tasks.postprocess.ConsolidateParentTractTask	Build a parent-only sibling table (object_parent) for blend-level diagnostics.
splitPrimaryObject lsst.pipe.tasks.split_primary.SplitPrimaryTask	Split object_all into object (primary detections) and object_nonprimary (tract-overlap duplicates).

Task	Brief description
metadetectionShear lsst.drp.tasks.metadetection_shear.MetadetectionShearTask	Run the metadetection weak-lensing measurement on the cell coadd, deriving a calibrated shear response per object.
consolidateShearObject lsst.pipe.tasks.postprocess.ConsolidateObjectTableTask	Combine per-patch shear catalogues into the tract-level object_shear_all.

Step 3a (continued) – tract-level analysis tasks.

Task	Brief description
analyzeObjectTableCore lsst.analysis.tools.tasks.ObjectTableTractAnalysisTask	Canonical metric & plot bundle on the tract-level object_all.
analyzeObjectParentTableCore lsst.analysis.tools.tasks.ObjectTableTractAnalysisTask	Same analysis bundle on the parent-only object table.
catalogMatchTract lsst.analysis.tools.tasks.astrometricCatalogMatch.AstrometricCatalogMatchTask	Match the tract object catalogue to an astrometric reference catalogue.
refCatObjectTract lsst.analysis.tools.tasks.refCatObjectAnalysis.RefCatObjectAnalysisTask	Plot astrometric residuals between the object catalogue and the reference catalogue.

Task	Brief description
photometricCatalogMatch lsst.analysis.tools.tasks.photometricCatalogMatch.PhotometricCatalogMatchTask	Match the tract object catalogue to a photometric reference catalogue.
photometricRefCatObjectTract lsst.analysis.tools.tasks.refCatObjectPhotometricAnalysis.RefCatObjectPhotometricAnalysisTask	Plot photometric residuals between the object catalogue and the reference.
plotPropertyMapTract lsst.analysis.tools.tasks.PerTractPropertyMapAnalysisTask	Render the per-tract HealSparse property maps as diagnostic figures.
makeBinnedCoaddImage lsst.analysis.tools.tasks.MakeBinnedCoaddTask	Bin the deep coadd down for visualisation.
makeWholeTractImage lsst.analysis.tools.tasks.WholeTractImageAnalysisTask	Render a whole-tract image of the binned deep coadd for QA.
makeBinnedTemplateNImage lsst.analysis.tools.tasks.MakeBinnedCoaddTask	Bin the template coadd's N-image (input-count map).

Task	Brief description
makeWholeTractTemplateNImage lsst.analysis.tools.tasks.WholeTractImageAnalysisTask	Render the whole-tract template N-image.
makeBinnedDeepNImage lsst.analysis.tools.tasks.MakeBinnedCoaddTask	Bin the deep-coadd N-image.
makeWholeTractDeepNImage lsst.analysis.tools.tasks.WholeTractImageAnalysisTask	Render the whole-tract deep-coadd N-image.
validateObjectTableCore lsst.analysis.tools.tasks.ObjectTableTractAnalysisTask	Column-validation checks on the assembled object table.

Step 3b – survey-wide coadd analysis (global).

Task	Brief description
consolidateHealSparsePropertyMaps lsst.pipe.tasks.healSparseMapping.ConsolidateHealSparsePropertyMapTask	Combine all per-tract HealSparse maps into one survey-wide map per property.
analyzeObjectTableSurveyCore lsst.analysis.tools.tasks.ObjectTableSurveyAnalysisTask	Survey-level analysis bundle on the consolidated object table.

Task	Brief description
makeMetricTableObjectTableCore lsst.analysis.tools.tasks.MakeMetricTableTask	Aggregate tract-level object metrics into a survey-wide metric table.
objectTableCoreWholeSkyPlot lsst.analysis.tools.tasks.WholeSkyAnalysisTask	Plot the object-table metrics across the whole sky.
makeMetricTableObjectParentTableCore lsst.analysis.tools.tasks.MakeMetricTableTask	Aggregate object-parent metrics across tracts.
objectParentTableCoreWholeSkyPlot lsst.analysis.tools.tasks.WholeSkyAnalysisTask	Plot the object-parent metrics across the whole sky.
makeMetricTableObjectTableCoreRefCatMatch lsst.analysis.tools.tasks.MakeMetricTableTask	Aggregate object-vs-refcat residual metrics across tracts.
objectTableCoreRefCatMatchWholeSkyPlot lsst.analysis.tools.tasks.WholeSkyAnalysisTask	Plot the object/refcat residual metrics across the whole sky.

Task	Brief description
plotPropertyMapSurvey lsst.analysis.tools.tasks.SurveyWidePropertyMapAnalysisTask	Render the survey-wide HealSparse property maps.
makeDeepCoaddInputSummary lsst.drp.tasks.make_coadd_input_summary.MakeCoaddInputSummary	Roll up the tract-level deep-coadd input summaries into one survey-wide table.
makeTemplateCoaddInputSummary lsst.drp.tasks.make_coadd_input_summary.MakeCoaddInputSummary	Same roll-up for the template coadd.

Steps 3c & 3d – per-band greyscale images and HiPS (global).

Task	Brief description
makeSingleBandRGBCoadd{U,G,R,I,Z,Y} lsst.pipe.tasks.prettyPictureMaker.PrettyPictureCoaddTask(perband)	Generate a stretched single-band greyscale visualisation of each band's deep coadd.
makeHighOrderHips{U,G,R,I,Z,Y} lsst.pipe.tasks.hips.HighResolutionHipsTask	Tile each band's deep coadd into the finest-resolution HiPS HEALPix tiles.
makeLowOrderHips{U,G,R,I,Z,Y} lsst.pipe.tasks.hips.GenerateHipsTask	Coarsen the high-order HiPS into the intermediate resolution levels.

Task	Brief description
writeAllSkyHipsInfo{U,G,R,I,Z,Y} lsst.pipe.tasks.hips.WriteAllSkyHipsTask	Write the HiPS metadata, all-sky preview and `properties` files for each band.

Stage 4 – Variability measurement

Step 4a – per-visit variability production (`[visit]`).

Task	Brief description
reprocessVisitImage lsst.drp.tasks.reprocess_visit_image.ReprocessVisitImageTask	Rebuild a final science-ready visit_image from post_isr_image, applying the refit PSF, new WCS, FGCM photo-calib and updated background.
standardizeSource lsst.pipe.tasks.postprocess.TransformSourceTableTask	Project the per-detector source catalogue from **reprocessVisitImage** into the canonical PreSource Parquet schema.
consolidateSource lsst.pipe.tasks.postprocess.ConsolidateSourceTableTask	Merge per-detector source tables into the visit-level source_all.
splitPrimarySource lsst.pipe.tasks.split_primary.SplitPrimaryTask	Split source_all into primary and non-primary detections per visit.
buildTemplate lsst.ip.diffim.getTemplate.GetTemplateTask	Carve and re-project the relevant patch of template_coadd onto the visit image's pixel grid.

Task	Brief description
subtractImages lsst.ip.diffim.subtractImages.AlardLuptonSubtractTask	PSF-match the template to the visit image (Alard & Lupton), subtract, and decorrelate the resulting difference image.
detectAndMeasureDiaSource lsst.ip.diffim.detectAndMeasure.DetectAndMeasureTask	Detect $\geq 5\sigma$ excursions in the difference image, measure them, and place sky sources for noise QA.
filterDiaSource lsst.ap.association.FilterDiaSourceCatalogTask	Apply preliminary cuts: remove sky sources and split off long-trailed (potentially streaking) sources.
computeReliability lsst.meas.transiNet.RBTransiNetTask	Run the RBTransiNet CNN on the science/template/difference triplet to assign each DIA source a real-bogus reliability score.
filterDiaSourcePostReliability lsst.ap.association.FilterDiaSourceReliabilityTask	Split DIA sources by reliability score into kept (high) and rejected (low) catalogues.
standardizeDiaSource lsst.ap.association.TransformDiaSourceCatalogTask	Project the high-reliability DIA sources into the canonical LSSTCam DIA schema.
consolidateVisitDiaSource lsst.pipe.tasks.postprocess.ConsolidateSourceTableTask	Merge per-detector DIA source tables into the per-visit dia_source_visit.
skyCorr lsst.pipe.tasks.skyCorrection.SkyCorrectionTask	Fit a focal-plane-wide background using all detectors in a visit, producing a sky-corrected visit mosaic.

Task	Brief description
analyzeRecalibratedStarObjectMatch lsst.analysis.tools.tasks.MatchedSourcesAnalysisTask	Match recalibrated stars against the deep-coadd object catalogue and analyse residuals.
analyzeSource lsst.analysis.tools.tasks.SourceTableVisitAnalysisTask	Run the canonical source-table analysis tools on source.
makeAnalysisSourceAstrometricRefMatch lsst.analysis.tools.tasks.astrometricCatalogMatch.AstrometricCatalogMatchVisitTask	Match per-visit sources to the astrometric reference catalogue.
analyzeSourceAstrometricRefMatch lsst.analysis.tools.tasks.refCatSourceAnalysis.RefCatSourceAnalysisTask	Plot astrometric residuals between sources and the reference catalogue.
makeAnalysisSourcePhotometricRefMatch lsst.analysis.tools.tasks.photometricCatalogMatch.PhotometricCatalogMatchVisitTask	Match per-visit sources to the photometric reference catalogue.

Task	Brief description
analyzeSourcePhotometricRefMatch lsst.analysis.tools.tasks.refCatSourcePhotometricAnalysis.RefCatSourcePhotometricAnalysisTask	Plot photometric residuals from the source/refcat photometric match.

Step 4b – Solar-System ephemerides (global).

Task	Brief description
generateEphemerides lsst.pipe.tasks.generateEphemerides.GenerateEphemeridesTask	Query the MPC (or local mirror) and write predicted positions of every known Solar System object for every visit.

Step 4c – per-tract variability assembly (`[tract]`).

Task	Brief description
associateDiaSource lsst.pipe.tasks.drpAssociationPipe.DrpAssociationPipeTask	Spatially link repeat DIA detections into DIA objects and attach matched SSO ephemerides.
calculateDiaObject lsst.pipe.tasks.drpDiaCalculationPipe.DrpDiaCalculationPipeTask	Compute per-DIA-object summary statistics (lightcurve moments, periodicity, colours).
consolidateDiaSource lsst.pipe.tasks.postprocess.ConsolidateTractTask	Combine per-patch DIA source tables into the tract-level dia_source.

Task	Brief description
consolidateDiaObject lsst.pipe.tasks.postprocess.ConsolidateTractTask	Combine per-patch DIA object tables into the tract-level dia_object.
forcedPhotObjectDetector lsst.pipe.tasks.forcedPhotDetector.ForcedPhotDetectorTask	Measure forced fluxes on visit_image at every object_patch position – the canonical object lightcurves.
standardizeObjectForcedSource lsst.pipe.tasks.postprocess.TransformForcedSourceTableTask	Project the forced-photometry output into the canonical object_forced_source_all schema.
splitPrimaryObjectForcedSource lsst.pipe.tasks.split_primary.SplitPrimaryTask	Split object_forced_source_all into primary and non-primary tables per patch.
forcedPhotDiaObjectDetector lsst.pipe.tasks.forcedPhotDetector.ForcedPhotDetectorTask	Forced photometry on visit_image at every dia_object position – the DIA-object lightcurves.
standardizeDiaObjectForcedSource lsst.pipe.tasks.postprocess.TransformForcedSourceTableTask	Project the DIA-object forced photometry into the canonical dia_object_forced_source schema.
associateAnalysisSource lsst.analysis.tools.tasks.AssociateSourcesTask	Build a tract-level associated-source catalogue for downstream analysis.

Task	Brief description
analyzeSourceAssociation lsst.analysis.tools.tasks.AssociatedSourcesTractAnalysisTask	Run association QA plots on the tract-level associated sources.
analyzeDiaSourceTableTract lsst.analysis.tools.tasks.DiaSourceTableTractAnalysisTask	Run the standard DIA-source analysis bundle on the tract-level dia_source.
makePrettyDirectWarp lsst.drp.tasks.make_direct_warp.MakeDirectWarpTask	Warp visit_images for the cosmetic colour-image release product, with sky correction applied.
makePrettyPsfMatchedWarp lsst.drp.tasks.make_psf_matched_warp.MakePsfMatchedWarpTask	PSF-match the pretty warps.
assemblePrettyCoadd lsst.drp.tasks.assemble_coadd.CompareWarpAssembleCoaddTask	Stack the pretty warps into the cosmetic pretty_coadd (no artefact masks written, SPIKE plane suppressed).
makeBinnedPrettyCoaddImage lsst.analysis.tools.tasks.MakeBinnedCoaddTask	Bin the pretty coadd for visualisation.
makeWholeTractPrettyCoaddImage lsst.analysis.tools.tasks.WholeTractImageAnalysisTask	Render the whole-tract pretty coadd.

Task	Brief description
makeBinnedPrettyNImage lsst.analysis.tools.tasks.MakeBinnedCoaddTask	Bin the pretty-coadd N-image.
makeWholeTractPrettyNImage lsst.analysis.tools.tasks.WholeTractImageAnalysisTask	Render the whole-tract pretty N-image.

Step 4d – survey-wide variability summaries (global).

Task	Brief description
makeAnalysisSourceAssociationMetricTable lsst.analysis.tools.tasks.MakeMetricTableTask	Aggregate tract-level source-association metrics into a survey-wide metric table.
makeAnalysisSourceAssociationWholeSkyPlot lsst.analysis.tools.tasks.WholeSkyAnalysisTask	Plot the source-association metrics across the whole sky.
consolidateSsTables lsst.pipe.tasks.consolidateSsTables.ConsolidateSsTablesTask	Fold per-visit Solar System detections into the canonical ss_source and ss_object tables using an HG12 phase-function fit.

Steps 4e & 4f – cosmetic RGB and EPO/release HiPS (global).

Task	Brief description
fixBackgroundPrettyCoadd lsst.pipe.tasks.prettyPictureMaker.FixBackgroundPrettyCoaddTask	Remove residual large-scale backgrounds from the pretty coadd.
fixStarsPrettyCoadd lsst.pipe.tasks.prettyPictureMaker.FixStarsPrettyCoaddTask	Repair saturated star cores in the pretty coadd to produce a clean colour image.
makeRGBPicturePrettyCoaddGRI lsst.pipe.tasks.prettyPictureMaker.PrettyPictureRGBTask	Combine the cosmetic g, r, i coadds into a colour image.
makeHighOrderHipsEpoGRI lsst.pipe.tasks.hips.HighResolutionHipsTask	Tile the EPO-purpose colour coadd into the finest HiPS tiles.
makeLowOrderHipsEpoGRI lsst.pipe.tasks.hips.GenerateHipsTask	Coarsen the EPO HiPS into intermediate resolution levels.
writeAllSkyHipsInfoEpoGRI lsst.pipe.tasks.hips.WriteAllSkyHipsTask	Write the EPO HiPS metadata/all-sky preview.
makeHighOrderHipsGRI lsst.pipe.tasks.hips.HighResolutionHipsTask	Tile the science-release multi-band colour coadd into the finest HiPS tiles.

Task	Brief description
makeLowOrderHipsGRI lsst.pipe.tasks.hips.GenerateHipsTask	Coarsen the release HiPS.
writeAllSkyHipsInfoGRI lsst.pipe.tasks.hips.WriteAllSkyHipsTask	Write the release HiPS metadata/all-sky preview.